

Minority Stakes, Technological Overlap, and Innovation*

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Abstract

Most competition regimes review only transactions that confer control. Thus, a firm can acquire a stake in a competitor without antitrust review. This paper studies how such acquisitions affect innovation when the two parties patent in the same technologies. Using worldwide patent, deal and financial data for 1997–2020, we compare acquirers and targets with matched control firms in a staggered difference-in-differences design. Within five years, cumulative patent output falls by 11.4% for acquirers and 4.8% for targets. Citation-weighted output falls by a similar margin. The decline is concentrated in the technology classes the two firms share, and it is absent when their technologies do not overlap. Targets respond only where the shared technologies matter to both parties and where the two firms also compete in product markets, while acquirers cut back regardless. These findings suggest that weakened competitive incentives account for the decline, rather than financial constraints or a division of research between the two parties.

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1 Introduction

When a firm acquires a minority stake in another firm, antitrust authorities have traditionally looked the other way. Most merger control regimes only review transactions that confer control, leaving acquisitions of less than 50% largely unscrutinized, even when the buyer and target are competitors.¹ Yet these acquisitions are common, accounting for roughly a quarter of global M&A activity by deal count,² and the literature has shown that they may soften competition, facilitate collusion, or lead to foreclosure.³ Their effect on innovation has received much less attention.

To see why a minority stake may matter for innovation, suppose a firm acquires a 20 percent stake in a rival that patents in the same technology. Every dollar of profit that the firm’s next innovation takes away from the rival now costs the acquirer 20 cents, so the return to innovating against the rival falls. The rival, in turn, faces a less aggressive competitor and adjusts its own research. Whether it cuts back as well or steps into the space the acquirer vacates depends on whether the two firms’ research efforts are strategic complements or substitutes. The sign of the acquirer’s response itself reverses if the two firms are complementors or trading partners rather than rivals, because then the acquirer’s innovation raises the target’s profit instead of lowering it. Which case prevails in actual minority stake acquisitions is an empirical question.

We assemble worldwide patent, deal, and financial data for 1997–2020. We focus on acquisitions in which the acquirer and the target were active in the same technology fields before the deal, where the acquirer’s innovation is most likely to come at the target’s expense. We measure this overlap with 4-digit IPC technology classes and call a deal an “overlapping minority stake acquisition” when the two firms share at least one such class among their pre-acquisition patents.⁴ We follow the patent output of acquirers and targets for five years after the deal. We measure it cumulatively, because annual patent counts are too volatile to read a firm’s research off a single year.

Firms involved in these acquisitions are not random. They are larger, more innovative and more liquid than other firms. We therefore compare each treated firm with a matched control that patents in the same technology classes, using a design built for staggered timing.

Overlapping minority stake acquisitions change both how much and where the involved firms innovate. Five years after the acquisition, treated firms have accumulated significantly less

¹Rock and Rubinfeld (2018) argue that the DOJ and the FTC generally have not challenged partial equity acquisitions below 20% absent evidence of control. Likewise, under the EU Merger Regulation, “When the acquisition of a minority shareholding is unrelated to acquisition of control, the Commission cannot investigate or intervene against it” (European Commission, 2014, para. 45). Minority stakes typically fall outside the scope of the regulation as Council Regulation (EC) No 139/2004, Art. 3(1)–(2) defines control as “the possibility of exercising decisive influence on an undertaking.”

²Calculations based on the Zephyr (Bureau van Dijk) database that we use in this paper.

³Shelegia and Spiegel (2024) provide an extensive literature review.

⁴IPC is a global hierarchical system used to categorize patents into specific technical fields. There are over 650 distinct 4-digit IPC codes (e.g., G06F for “Data Processing”), which provide a granular measure of technological proximity.

patent output than their matched controls, 11.4% less for acquirers and 4.8% less for targets. The gap opens in the first year after the acquisition and widens steadily thereafter. These effects arise at ownership levels well below a majority, and most of the acquisitions we study leave the acquirer holding less than a quarter of the target. Citation-weighted patent output declines by a similar margin, so the cutback is not confined to marginal, low-value patents. The decline is also specific to the technological relationship between the parties. After minority acquisitions between firms whose technologies do not overlap, we find no comparable decline. The pre-acquisition coefficients are close to zero throughout, and the estimates survive alternative definitions of overlap, of the control group and of the sample of deals, including the exclusion of Japan and Korea.

If the acquirer internalizes part of the harm its innovation inflicts on the target, the decline should be concentrated where the two firms' technologies overlap, not spread across their portfolios. When we split each firm's patenting into the classes it shares with its counterpart and all other classes, the decline is concentrated in the shared classes, for acquirers and targets alike. The classes in which one party cuts back most are the classes in which the other does too. The target's response is strongest in acquisitions in which the shared fields are central to both parties' research and in acquisitions in which the two firms also compete in product markets, which is where a reaction to the acquirer's cutback should be expected.⁵

Taken together, these patterns point to internalization. Once the acquirer holds a stake in a rival, it cuts its investment in the technologies they share, and the target, whose returns in these technologies depend on the acquirer's effort, follows. Firm-wide explanations such as financial constraints or managerial distraction, the elimination of duplicate research efforts and a governance channel through board seats each fit part of this pattern. None of them explains all of these patterns together.

Recently, policymakers have started to be concerned with minority shareholdings. For instance, the 2023 FTC-DOJ Merger Guidelines now state that "cross-ownership and common ownership can reduce competition by softening firms' incentives to compete, even absent any specific anticompetitive act or intent." Moreover, the guidelines warn that a partial owner "may decide not to develop a new product feature to win market share from the firm in which it has acquired an interest, because doing so will reduce the value of its investment in its rival."⁶

Enforcement has taken several different forms. One of the first prominent antitrust actions against a minority shareholding came in 2013, when the UK Competition Commission required

⁵Because our overlap definition counts any shared class, the average target effect understates the response of firms whose research programs genuinely collide.

⁶Both quotations are from Guideline 11 of the U.S. Department of Justice and Federal Trade Commission (2023), the second from its Section 2; available at <https://www.justice.gov/atr/merger-guidelines/applying-merger-guidelines/guideline-11>.

Ryanair to reduce its 29.8% stake in Aer Lingus to 5%, having found that the stake was expected to substantially lessen competition in airline services. In 2021, the DOJ settled with Geisinger over its planned 30% investment in a rival Pennsylvania hospital, limiting the stake to 7.5% as a passive investment. The FTC has opened an investigation into Microsoft’s investment in OpenAI.

In June 2025, the European Commission fined Delivery Hero and Glovo a total of € 329 million for participating in a cartel in the online food delivery sector, after Delivery Hero had acquired a non-controlling stake in Glovo in 2018, the first time the Commission sanctioned the anti-competitive use of a minority share in a competing business. Two months later, the Commission conditioned its approval of Naspers’ acquisition of Just Eat Takeaway.com on the divestiture of Prosus’s 27.4% stake in competitor Delivery Hero, finding that the structural link could reduce Just Eat’s incentive to compete.⁷ The Delivery Hero case thus turned on coordinated effects, the Naspers case on unilateral effects.

The closest empirical study to ours is Bostan and Spatareanu (2018), who find that minority acquisitions boost target innovation, an effect driven by financial transfers to constrained firms rather than by competitive interaction. To our knowledge, no prior study examines how minority stake acquisitions affect innovation when the parties operate in the same technology space. Reviewing every minority investment would be costly, and technological overlap between the parties identifies the transactions in which innovation incentives are most likely to change.

The rest of the paper is organized as follows. Section 2 discusses related literature. Section 3 presents our theoretical framework. Section 4 presents our data and Section 5 discusses the empirical strategy. Section 6 reports the results together with the robustness analyses. Section 7 concludes.

2 Related Literature

If a firm owns part of its competitor, aggressive pricing or innovation hurts the value of that stake. The firm has less reason to compete. Reynolds and Snapp (1986), building on Bresnahan and Salop (1986), showed that even small stakes lead to lower output and higher prices in models of oligopoly with partial ownership. The result extends to passive stakes under a range of market structures (Farrell and Shapiro, 1990; Shelegia and Spiegel, 2012; Jovanovic and Wey, 2014) and to coordinated effects: structural links between competitors can stabilize tacit collusion by making

⁷Case AT.40700 (European Commission, 2025) and Case COMP/M.11936 (Paul, Weiss, Rifkind, Wharton & Garrison LLP, 2025). Prosus was also required to waive its voting rights, board representation, and information rights even after the sell-down. For the other cases in this and the preceding paragraph, see <https://www.gov.uk/cma-cases/ryanair-aer-lingus-merger-inquiry> (Ryanair/Aer Lingus), <https://www.fiercehealthcare.com/hospitals/doj-reaches-settlement-geisinger-health-evangelical-antitrust-case> (Geisinger), <https://www.nytimes.com/2024/01/26/business/dealbook/ftc-ai-deals-microsoft-openai.html> (Microsoft/OpenAI), and https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1356 (Delivery Hero/Glovo).

deviation less attractive (Malueg, 1992; Gilo et al., 2006). Empirically, Nain and Wang (2018) estimate that horizontal minority acquisitions in U.S. manufacturing raised prices by roughly 2% and increased margins by 0.7%. Dietzenbacher et al. (2000) find elevated margins among Dutch financial firms with cross-shareholdings. Heim et al. (2022) document a spike in minority stake acquisitions following the introduction of leniency programs, suggesting that firms use these stakes to reinforce cartels rather than as passive investments.

Whether minority stakes increase or decrease innovation depends on the balance between two forces. Owning part of a rival softens the incentive to compete through R&D, but it can also help internalize technology spillovers that would otherwise be free-ridden on. López and Vives (2019) and Stenbacka and Van Moer (2023) show that the net effect hinges on the level of spillovers and on market structure: when spillovers are low, the competition-softening channel dominates; when they are high, internalization can outweigh the loss of competitive pressure. Shelegia and Spiegel (2024) show that the sign further depends on the symmetry of stakes and the cost of innovation. The parameter values that tip the balance differ by industry and deal structure, so the question is ultimately empirical.

A large literature studies the effect of full mergers on innovation. In the context of M&A, the “innovation theory of harm” holds that mergers reduce innovation by internalizing the business-stealing externality (Federico et al., 2017), though mergers can also facilitate knowledge-sharing (Denicolò and Polo, 2019; Bourreau and Jullien, 2018). Marshall and Parra (2019) show that competition affects innovation through both the number of firms racing to innovate and the profit gap between leaders and followers, and that these channels can push in opposite directions. Empirically, Haucap et al. (2019) find that pharmaceutical mergers reduce innovation, with effects concentrated in R&D-intensive markets. Cunningham et al. (2021) identify “killer acquisitions” in pharmaceuticals, cases where incumbents acquire targets specifically to discontinue overlapping projects. Malek et al. (2025) add nuance, finding positive effects when incumbents acquire early-stage projects similar to their own. Bena and Li (2014) show that acquirers with prior technological linkage to their targets produce more patents after the deal. Full acquisitions, though, allow the acquirer to integrate the target’s R&D operations and realize synergies. Minority stakes provide the financial link without the integration, leaving the competition-softening channel largely unopposed.

A growing literature examines how common ownership by institutional investors affects innovation (e.g., Antón et al., 2023, 2024; Li et al., 2023; Lindsey, 2008; Kostovetsky and Manconi, 2020; Borochin et al., 2020). The channels differ from direct corporate minority stakes. Common ownership is indirect and typically passive. The mechanism runs through portfolio incentives of diversified investors, not through strategic interaction between rival firms. Corporate minority

stakes create sharper incentive distortions because the acquirer has a direct financial interest in its competitor's profits and, in many cases, access to its governance.

The closest empirical study is Bostan and Spatareanu (2018), who find positive innovation effects for U.S. targets of minority equity purchases. Their mechanism is financial, since they view minority stakes as a funding bridge for constrained firms, without restricting attention to horizontal deals. The present paper isolates acquisitions between firms that work in the same technology fields, where the motive is more likely strategic than financial. None of the prior studies cited above examines what happens to innovation when a firm acquires a partial stake in a firm whose technologies overlap with its own.

3 Theory

We develop a theoretical framework that establishes the conditions under which an increase in the stake of one firm in another will strengthen or weaken the incentives of the acquirer and target to invest in R&D. As we will show below, the conditions depend critically on two factors: whether the acquirer's R&D investment imposes a positive or a negative externality on the target and whether the R&D investments of the acquirer and target are strategic substitutes or strategic complements.⁸

Consider a duopoly where one firm (an acquirer) acquires a minority stake α in the other firm (the target firm). Let x^a and x^t be the R&D investments of the acquirer and target and let their corresponding profit functions be $\pi^a(x^a, x^t)$ and $\pi^t(x^a, x^t)$. Assuming that the acquisition does not involve control, each firm chooses its investment in R&D to maximize its overall profit. The acquirer chooses x^a to maximize

$$\Pi^a(x^a, x^t, \alpha) = \pi^a(x^a, x^t) + \alpha \pi^t(x^a, x^t), \quad (1)$$

and the target chooses x^t to maximize

$$\pi^t(x^a, x^t). \quad (2)$$

A Nash equilibrium in the R&D game, $(\hat{x}^a, \hat{x}^t)$, is given by the solution to the following first-order conditions:

$$\Pi_a^a(x^a, x^t, \alpha) = \pi_a^a(x^a, x^t) + \alpha \pi_a^t(x^a, x^t) = 0, \quad \text{and} \quad \pi_t^t(x^a, x^t) = 0, \quad (3)$$

⁸Aoki and Spiegel (2009) show that when R&D competition is followed by competition in the product market, R&D investments may be either strategic substitutes or strategic complements, depending on whether it is more important for firms to take the technological lead or to catch up in case another firm takes the technological lead.

where subscripts denote partial derivatives. The expression $\pi_a^t(x^a, x^t)$ represents the externality that the acquirer's R&D investment imposes on the target. In general, the acquirer's R&D investment boosts the acquirer's sales. If the target sells a complementary product or is a trading partner of the acquirer (either a customer or a supplier), it benefits from the increase in the acquirer's sales, i.e., $\pi_a^t(x^a, x^t) > 0$. By contrast, if the target is a rival, the increase in the acquirer's sales harms the target, i.e., $\pi_a^t(x^a, x^t) < 0$.

We will assume that the second-order conditions are satisfied, i.e., $\Pi_{aa}^a(x^a, x^t, \alpha) < 0$ and $\pi_{tt}^t(x^a, x^t) < 0$, and will assume in addition that the following inequalities, which ensure that the Nash equilibrium is unique and stable (see Shapiro, 1989), hold:

$$\Pi_{aa}^a(x^a, x^t, \alpha) + \left| \Pi_{at}^a(x^a, x^t, \alpha) \right| < 0, \quad \text{and} \quad \pi_{tt}^t(x^a, x^t) + \left| \pi_{at}^t(x^a, x^t) \right| < 0. \quad (4)$$

The first-order conditions implicitly define the best-response functions of the two firms. Using the first-order conditions, the slopes of the best-response functions in the (x^a, x^t) space are given by:

$$\frac{\partial x^a}{\partial x^t} = -\frac{\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha)}{\Pi_{aa}^a(\hat{x}^a, \hat{x}^t, \alpha)}, \quad \text{and} \quad \frac{\partial x^t}{\partial x^a} = -\frac{\pi_{at}^t(\hat{x}^a, \hat{x}^t)}{\pi_{tt}^t(\hat{x}^a, \hat{x}^t)}. \quad (5)$$

The strategies of the acquirer and target are strategic substitutes if the slopes are negative and are strategic complements if they are positive. Given the second-order conditions, the strategies of the acquirer and target are strategic substitutes if $\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) < 0$ and $\pi_{at}^t(\hat{x}^a, \hat{x}^t) < 0$, and are strategic complements if $\Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) > 0$ and $\pi_{at}^t(\hat{x}^a, \hat{x}^t) > 0$.

We now examine how the acquirer's stake in the target firm, α , affects the R&D investments. To this end, we fully differentiate the first-order conditions and divide through by $\partial \alpha$. The resulting comparative matrix is given by

$$\begin{pmatrix} \Pi_{aa}^a(\hat{x}^a, \hat{x}^t, \alpha) & \Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha) \\ \pi_{at}^t(\hat{x}^a, \hat{x}^t) & \pi_{tt}^t(\hat{x}^a, \hat{x}^t) \end{pmatrix} \begin{pmatrix} \frac{\partial \hat{x}^a}{\partial \alpha} \\ \frac{\partial \hat{x}^t}{\partial \alpha} \end{pmatrix} = \begin{pmatrix} -\pi_a^t(x^a, x^t) \\ 0 \end{pmatrix}. \quad (6)$$

Solving (6), yields

$$\frac{\partial \hat{x}^a}{\partial \alpha} = -\frac{\pi_a^t(x^a, x^t) \times \pi_{tt}^t(x^a, x^t)}{\Delta}, \quad \text{and} \quad \frac{\partial \hat{x}^t}{\partial \alpha} = \frac{\pi_a^t(x^a, x^t) \times \pi_{at}^t(x^a, x^t)}{\Delta}, \quad (7)$$

where $\Delta = \Pi_{aa}^a(\hat{x}^a, \hat{x}^t, \alpha)\pi_{tt}^t(\hat{x}^a, \hat{x}^t) - \Pi_{at}^a(\hat{x}^a, \hat{x}^t, \alpha)\pi_{at}^t(\hat{x}^a, \hat{x}^t)$ is the determinant of the comparative statics matrix. By (4), $\Delta > 0$.⁹

⁹ $\Delta > 0$ ensures that the Nash equilibrium is stable in the sense that the best-response function of the acquirer crosses the best-response function of the target firm in the (x^a, x^t) space from above if strategies are strategic substitutes and cuts it from below if the strategies are strategic complements.

Equation (7) has three implications. First, an increase in α does not affect the R&D investments of either the acquirer or the target if $\pi_a^t(x^a, x^t) = 0$, i.e., the acquirer's R&D investment does not impose an externality on the target. Second, given that $\pi_{tt}^t(x^a, x^t) < 0$, an increase in α boosts the acquirer's R&D investment if $\pi_a^t(x^a, x^t) > 0$, i.e., the externality is positive, but decreases it if the externality is negative, i.e., $\pi_a^t(x^a, x^t) < 0$. Third, the target reacts in the same direction as the acquirer if R&D investments are strategic complements, i.e., $\pi_{at}^t(x^a, x^t) > 0$, and reacts in the opposite direction if strategies are strategic substitutes, i.e., $\pi_{at}^t(x^a, x^t) < 0$.

In terms of our setting, these implications yield three predictions. First, an acquirer facing a negative externality reduces its R&D investment (P1). Second, the target moves in the same direction if R&D investments are strategic complements (P2). Third, absent an externality, which in our setting means absent technological overlap, neither side responds (P3).

4 Data

4.1 Data Sources

We measure innovation with PATSTAT, the patent database of the European Patent Office. It records patent applications worldwide, and its coverage reaches back to the nineteenth century. For every granted patent we extract the earliest filing year, the forward citations it later receives, and the 4-digit IPC classes the invention falls into. To place each patent in the year the research was done, we date it by filing and not by grant.¹⁰ This gives two measures of innovation output for each firm-year, a count of patents and a count of citations, and it gives the set of technology classes in which a firm is active. The unit of observation is the patent application. A firm's citation count in a year is the number of citing patent families, summed over the applications it filed in that year. Citations are a stock at the date of extraction rather than a flow, so each patent carries the citations it had attracted by 2024 and is attributed to the year it was filed.

A patent application is filed once an invention is judged protectable, long before it reaches a market. Our counts therefore capture research output at the point of invention. They capture neither research effort nor commercial success. The downside is that not every invention is patented, and firms differ in how far they rely on formal protection.

Patent counts also weight a trivial filing and a breakthrough equally. We therefore measure output in a second way, by the citations a firm's patents attract, because high-value inventions are cited more extensively than low-value ones (Harhoff et al., 1999). Citations are also far more unequal than patents. The most cited tenth of the patents in our data accounts for close to

¹⁰We keep only applications that had been granted when the data were extracted in 2024. The extract carries grant status but no grant date, so a patent enters a firm's pre-acquisition stock if it was filed before the acquisition year and had been granted by 2024.

sixty percent of all citations.¹¹ The two measures come apart when the composition of a firm’s patenting changes rather than its amount.

Our deal data come from Bureau van Dijk’s Zephyr database, which records completed transactions worldwide from 1997 to 2020, with completion dates, the stake acquired, and the identities of both parties. We keep deals in which the reported final stake stays below 50%. In these transactions the target holds at least one granted patent in 41% of cases, and the acquirer or one of its corporate shareholders in 33%.¹²

Firm characteristics come from Bureau van Dijk’s Orbis database, which reports balance sheet and profit-and-loss information for approximately 45 million companies worldwide. We merge the three sources at the firm- year level. From Orbis we take the variables on which treated and control firms are later matched, among them sales, total assets, liquidity ratios and employment. Only about 5% of firms in Orbis hold a patent at all (Gugler et al., 2023). Firms on either side of a minority stake acquisition are therefore six to eight times as likely to patent as the typical firm.

4.2 Identifying Overlapping Minority Stake Acquisitions

Zephyr records 178,727 completed minority stake acquisitions in which both parties are identified. Three groups of deals then leave that set. To restrict the sample to acquisitions of one firm by another, we drop share buybacks and self-tenders. To exclude acquirers with no research of their own onto which a target’s innovation could spill, we drop deals in which either party’s primary industry code identifies it as an Investor or a Management investment office.¹³ And to keep only pairs whose technological relationship can be established, we drop transactions in which neither side has patent data. This leaves 24,267 minority stake acquisitions.

A minority stake changes innovation incentives through the externality the acquirer’s research imposes on the target. That externality is largest where the two firms work on the same technologies. We call a deal an overlapping minority stake acquisition when the acquirer, or its immediate parent, and the target share at least one 4-digit IPC class.¹⁴ To keep the classification free of anything the deal itself caused, we count only classes that appear among patents applied for before the acquisition year and granted later. By this definition, 56% of the deals overlap, covering 4,561 targets and 2,898 acquirers.

Industry codes are the obvious alternative. An industry code records what a firm sells, while an IPC class records what it researches. The externality originates in research, which is why we classify horizontality by technology class, and we report a version of the analysis that classifies

¹¹Patents filed between 1985 and 2020. The median patent is cited five times and the mean sixteen.

¹²Shares across 178,523 completed minority stake acquisitions with both parties identified, after removing share buybacks and self-tenders. Both parties hold a patent in 17% of these deals.

¹³SIC codes 6799 and 6722, respectively.

¹⁴We use the immediate parent rather than the ultimate corporate owner, since ultimate owners are often diversified conglomerates whose broad patent portfolios would generate spurious overlapping classifications.

it by industry code.¹⁵ An obvious drawback is that overlap can only be computed where both parties patent, which is what restricts the sample.

4.3 Treated Sample

We analyze targets and acquirers separately, in two parallel datasets.¹⁶ So that the estimates are not confounded by outright takeovers, we first exclude deals in which the acquirer obtains majority control of the target within five years.¹⁷ We then drop firms for which Orbis reports no financial data in the year of the acquisition, and firms that are not corporate entities. Finally we require three consecutive years of patent data, so that a firm's patenting can be followed over time rather than observed once. This leaves 2,682 targets and 1,239 acquirers.

Taken together, these requirements select firms that are larger and more research-intensive than the average party to a minority stake acquisition. That is the price of observing a patent trajectory on both sides of a transaction. What the restrictions buy is a technological relationship that can be measured, for every firm in the sample, before the deal takes place.¹⁸

4.4 Descriptive Statistics

These firms are headquartered in 40 countries, and most deals are domestic.¹⁹ Japan and South Korea account for the largest shares of them.²⁰ Both countries have an institution that produces such holdings by design, though not the same one. Japanese *keiretsu* are groups of peer firms, often centered on a main bank, that hold small stakes in one another reciprocally as a standing governance arrangement. Korean *chaebol* are controlled by a founding family, which holds the group together through a chain of stakes across affiliates rather than through mutual holdings. In both cases the stakes are held for decades, and neither is a one-off transaction. The sample is concentrated in manufacturing and, to a much smaller extent, in information industries.²¹

Acquirers tend to be substantially larger than targets. At the median, a target has about one tenth of the sales and total assets of its acquirer, and an acquirer roughly four times the sales and total assets of its target.²²

¹⁵Figure A.XV uses 4-digit NAICS, NACE and SIC codes.

¹⁶Table A.I in the Appendix reports how many firms remain after each of the steps below.

¹⁷This exclusion conditions the sample on a post-acquisition event. Section 6.7 shows that the results are essentially unchanged when these deals are retained and the affected firms are instead censored from the year the stake turns into a majority position.

¹⁸Appendix A.1 compares the treated sample with the full set of overlapping deals.

¹⁹At their first overlapping acquisition, 68% of targets were acquired by a firm in the same country, and 80% of acquirers acquired a domestic target.

²⁰32% and 16% respectively.

²¹68% and 6% respectively. Detailed breakdowns by year, country and industry appear in Figures A.I–A.III in the Appendix.

²²Table A.IV in the Appendix. The two panels condition on different subsets of deals, those for which the counterparty's financial data are available in our panel, which is why the two median ratios are not exact inverses of each other. Table A.III provides further pre-acquisition summary statistics.

Figure I characterizes the acquisitions along the two dimensions that carry the analysis, the size of the stake and the degree of technological overlap. The stakes are small. The median acquirer buys 10% of its target, the median target sells 5% of its shares, and most acquisitions lie below 25% (panel a). A typical transaction is a larger firm taking a few percent of a smaller firm working on the same technologies, far from anything that resembles control.²³

Technological overlap, by contrast, varies enormously (panel b). For each treated firm we compute the share of its pre-acquisition patents that falls into the 4-digit IPC classes in which its counterpart also patents. Consider a target with 100 patents whose acquirer is active in two of the target's classes, say vehicles and electrical machinery. If 20 of the target's patents belong to those two classes, its overlap share is 20%. The median is 25% for targets and 38% for acquirers.

Further, the distribution has two peaks. Almost a third of the targets share only a marginal part of their portfolio with the acquirer, a few percent of their patents. For nearly a fifth, virtually all of their patents lie in shared classes, so that their entire technology field is also covered by the acquirer. Our definition, which requires one shared class, therefore spans both firms whose research programs barely touch and firms that work on exactly the same problems.

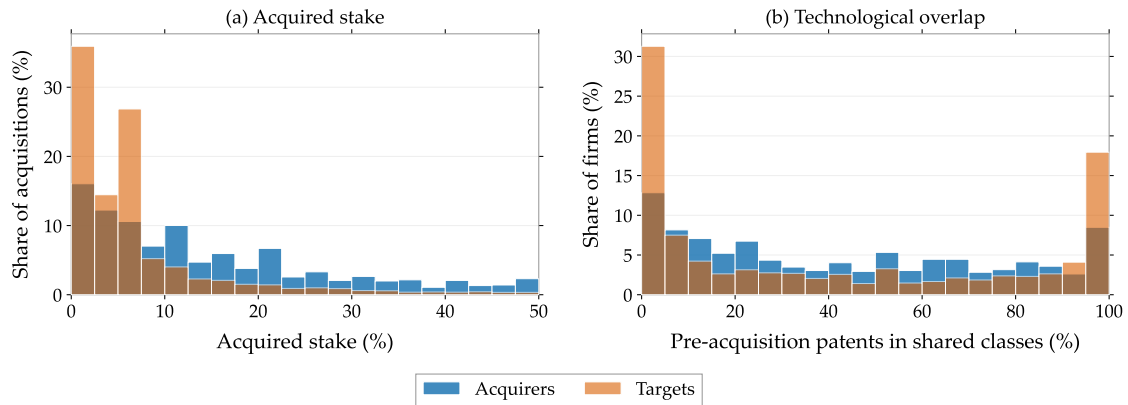


Figure I: Acquired Stakes and Technological Overlap in the Matched Sample

Panel (a) shows the distribution of the acquired stake across the overlapping minority stake acquisitions of treated acquirers and targets in the matched sample; each bar covers 2.5 percentage points. Panel (b) shows, for each treated firm whose counterpart can be identified in the deal data, the share of its pre-acquisition patents that falls into the 4-digit IPC classes in which its counterpart also holds patents; each bar covers 5 percentage points. Medians are 10% (acquirers) and 5% (targets) for the stake and 38% and 25% for the overlap share.

Finally, more than a third of the sample firms are involved in more than one overlapping minority acquisition during the observation period, sometimes as acquirer and sometimes as target.²⁴ Innovation activity is concentrated: the top 15% of firms in our sample account for

²³The distribution differs between the acquirer and target samples because some acquisitions are reported in one sample but not in the other due to data availability. Financial or patent data might be available for the acquirer but not the target, or the reverse.

²⁴Approximately 28% of the targets were subject to successive stake acquisitions or separate transactions over time by the same acquirer; about 17% of acquirers acquired stakes in the same target more than once. In addition, roughly 22% of sample firms were also involved in non-overlapping minority stake acquisitions.

more than 90% of the total patent stock. That concentration is a feature of global research and not of our sample. Among the largest research spenders in the world, the top 15% already account for nearly three quarters of total spending.²⁵

5 Research Design

Firms involved in overlapping minority stake acquisitions are larger, more innovative and more liquid than other firms (Figure A.IV in the Appendix), and the acquisitions occur at different times. Annual patent counts are also volatile. For the median firm the standard deviation of the annual count is as large as its mean, so a single year's count is a noisy measure of research output. We therefore combine propensity score matching with a staggered difference-in-differences estimator, and measure patent output cumulatively.

5.1 Matching Procedure

The control group consists of corporate Orbis firms that never take part in a minority stake acquisition. Overlap requires shared technology classes, so treated firms patent by definition. We therefore restrict the pool to firms with five consecutive years of patent and financial data, which the matching then uses to compare pre-acquisition trajectories. This leaves more than 92,000 potential controls.

From this pool we select controls by propensity score matching (Rosenbaum and Rubin, 1983, 1985). A probit model predicts involvement in an overlapping minority stake acquisition from sales, employment, total assets, intangible assets, the liquidity ratio and age in the year before the deal. Two firms can hold the same patent stock in the year before a deal and have reached it from opposite directions. The score therefore also includes the levels of cumulative patents and cumulative citations at $t - 1$, $t - 2$ and $t - 3$.²⁶ The model also carries country, year and 2-digit industry fixed effects.²⁷

Each treated firm is then matched to the control firm with the closest propensity score, subject to three restrictions. Treated and control firms must be observed in the same year, which keeps the comparison inside a single filing environment. Controls must hold at least one patent in a 4-digit IPC class in which the treated firm patented before the acquisition, which ensures that matched pairs operate in the same technology domain. We restrict the analysis to firms that fall

²⁵Based on the 2022 EU Industrial R&D Investment Scoreboard, the top 15% of the 2,500 largest R&D investors account for nearly 75% of total Scoreboard R&D expenditure, and this within a group that is already restricted to the world's largest R&D spenders. Measured over a broader population such as ours, concentration is naturally even higher. See <https://iri.jrc.ec.europa.eu/scoreboard/2022-eu-industrial-rd-investment-scoreboard>.

²⁶Table A.II in the Appendix reports the estimates.

²⁷Countries differ in how readily firms patent at all, years differ in filing conditions, and industries differ in how much there is to invent.

within the common support. Treated firms whose propensity score lies above anything available in the control pool are dropped.²⁸

One concern is that the technology restriction admits controls the acquisition itself may reach. A control patenting in the same classes as a treated firm is affected when the acquirer cuts back in those classes, in which case the comparison understates the effect. We therefore re-match treated firms to controls that share a 3-digit class but hold no patent in any of the treated firm's 4-digit classes, and the acquirer estimate is essentially unchanged.²⁹

We match with replacement and weight each control by the number of treated firms it serves. Although matching without replacement is cleaner, it leaves one in five treated targets without a match. We report both.³⁰

Two calipers then discard matches that the propensity score alone would accept. We drop pairs whose scores differ by more than 0.05, and we require treated and control firms to have similar pre-acquisition growth in cumulative patent and citation output, allowing at most 0.1 between their five-year log slopes. Both calipers tighten the comparison at the cost of treated firms.³¹

Taken together, the restrictions keep most of the treated firms. About three in four find a match on either side.³²

Before matching, treated firms are far larger and far more patent-intensive than the typical potential control. Matching closes most of that gap. The matched controls are of similar size and age and, if anything, hold somewhat larger patent and citation stocks.³³

However, the balance figure cannot settle whether the matching has lined up trajectories. It compares levels at a single date. If it has not, we should see treated and control firms drifting apart before the acquisition, and the pre-acquisition coefficients of the event study would show it.

5.2 Staggered Difference-in-Differences

Under staggered timing, two-way fixed effects would use firms that are already treated as controls for firms treated later. We therefore estimate the event study with the de Chaisemartin and D'Haultfoeuille (2026) estimator, which in each period compares newly treated firms only with firms that are not yet treated. This avoids the negative weighting the comparison can otherwise

²⁸Twelve targets and eleven acquirers.

²⁹Section 6.7 reports this check. For targets the smaller pool of admissible controls leaves the estimate imprecise.

³⁰Figure A.X in the Appendix.

³¹The same calipers are applied to the non-overlapping acquisitions that serve as a placebo in Section 6.4, which show no decline.

³²Of the 2,682 targets and 1,239 acquirers that enter the procedure, 2,205 and 1,033 have complete pre-acquisition covariates, and 2,069 and 919 find a match, drawing on 1,304 and 666 distinct controls.

³³Standardized biases exceed 100% for size and innovation stocks before matching and fall by more than 90%, with a largest remaining bias of 12%. The pseudo- R^2 of the probit re-estimated on the matched sample falls from 0.41–0.45 to 0.05–0.08. Figure A.IV in the Appendix reports the comparison, and Figures A.V and A.VI the patent distributions.

produce.³⁴

The dependent variable is the log of cumulative granted patents, or of cumulative forward citations, plus one.³⁵ Cumulative output is what determines a firm’s technological position, which rests on the stock of research a firm has built up.³⁶ Differences in a cumulative measure are permanent. A single weak year therefore appears at every later horizon. We check the baseline against the same design estimated on annual production, scaled by the firm’s pre-acquisition patent stock.³⁷

Let $\tau \in \{-5, \dots, 5\}$ denote event time relative to the last year before a firm’s first overlapping minority stake acquisition. That year serves as the reference period, $\tau = 1$ is the acquisition year, and $\tau = 5$ is the fifth year of exposure. The estimating equation is given by

$$\ln(\text{Patent}_{i,t} + 1) = \gamma_i + \gamma_t + \sum_{\substack{\tau=-5, \\ \tau \neq 0}}^5 \beta_\tau D_{i,t}^\tau + u_{i,t} \quad (8)$$

where $D_{i,t}^\tau$ denotes an indicator for year t lying τ years from firm i ’s reference period, γ_i denotes firm fixed effects and γ_t calendar-year fixed effects. Each β_τ is the difference in cumulative patent output between treated firms and their matched controls at horizon τ . Standard errors are clustered at the firm level.

We follow the acquisitions for five years. Beyond that the sample thins. Our baseline estimand is the gap at the fifth year. The identifying assumption is that, absent the acquisition, treated firms’ cumulative patenting would have followed the path of their matched controls, conditional on the characteristics the propensity score balances.

Matching conditions on observables. It cannot rule out that treated and control firms differ in time-varying ways we do not observe, and the firm fixed effects absorb such differences only where they are constant over time. Flat pre-acquisition coefficients are evidence for the assumption but do not establish it, so we also report the size of violation that would overturn the results.³⁸

³⁴See, e.g., de Chaisemartin and D’Haultfoeuille (2026), de Chaisemartin and D’Haultfoeuille (2020), Borusyak et al. (2024), Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021). Roth et al. (2023) recommend a staggered-robust estimator for event-study plots on this ground.

³⁵We count cumulative granted patents, regardless of whether they already expired or were not renewed. We use $\ln(x + 1)$ for patents and citations for consistency across outcomes. Results for cumulative patents are similar using $\ln(x)$, with year-five effects of -12.7% for acquirers and -5.3% for targets.

³⁶Cumulative and fixed multi-year patent outcomes are also used by Bena and Li (2014), Coelli et al. (2022), and Xie and Zhou (2026).

³⁷Section 6.

³⁸Section 6.2, following Roth et al. (2023) and Rambachan and Roth (2023).

6 Results

6.1 Main Effects on Patent Output

Figure II reports the estimated effect on cumulative patent output for each of the five years before and after the acquisition. Acquirers are in the left panel, targets in the right. The horizontal axis is time in years relative to the last year before the acquisition, which is normalized to zero. Each point is the difference in cumulative patent output between treated firms and their matched controls. Because the dependent variable is in logs, the figures report percentages, $100 \times (\exp(\beta_\tau) - 1)$.

Before the acquisition, the coefficients are small and statistically indistinguishable from zero, indicating no differential pre-trends between treated firms and the control group.³⁹ After the acquisition the two groups separate, and the gap widens each year. Within five years acquirers have accumulated 11.4% fewer patents than their controls, and targets 4.8% fewer.⁴⁰ The acquirer's decline is the larger of the two. In the theory the acquirer responds to the stake it now holds, while the target responds only to the change in the acquirer's behavior.

These are effects on accumulation, not on the level of the stock. Over the same five years the control group adds about 60% to its cumulative patent output on the acquirer side and about 45% on the target side.⁴¹ Treated firms keep patenting, only more slowly.

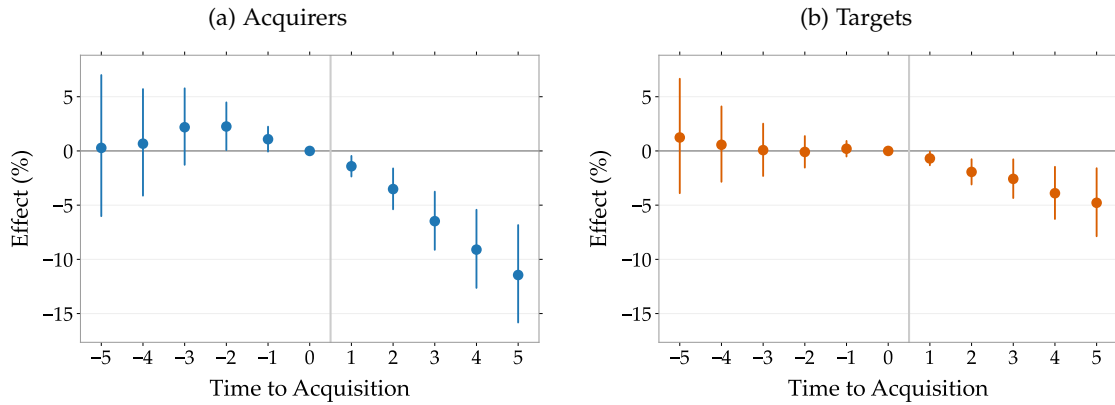


Figure II: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. The event-study coefficients are invariant to the treatment coding of Table I (binary or discrete). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

³⁹Joint tests that all five placebo coefficients are zero do not reject, with $p = 0.86$ for targets and $p = 0.19$ for acquirers.

⁴⁰These estimates are robust to standard variations of the matching procedure and to restricting exposure to the first acquisition (Figures A.IX–A.XI and A.XIV in the Appendix). The target estimate averages over deals in which the shared technology classes are central to both parties and deals in which they are not. Section 6.6 reports the split.

⁴¹Median five-year growth in cumulative patents among matched controls, 61.7% and 44.1%.

Table I reports the average effect over the full five-year window. Because that average includes the early years, when the cumulative gap is still small, the magnitudes are smaller than the year-five endpoints. Treating each acquisition as a single event gives an average decline of about 6% for acquirers and 3% for targets. Letting repeated acquisitions accumulate gives about 5% and 2%. The two codings agree.⁴²

Annual patent output shows the decline directly. We estimate the same design on yearly granted patents, scaled by the firm's pre-acquisition patent stock. Output is lower throughout the post-acquisition window, at about 7 patents per 100 pre-acquisition patents on both sides in the fifth year.⁴³ Non-overlapping acquisitions show no comparable decline.

Neither coding distinguishes a stake of two percent from one of forty. The distinction should matter. If the acquirer holds back because it now shares in the profit its rival earns, small stakes should change little. We therefore split the acquisitions at a stake of 10%, and compare each group to its own controls.⁴⁴

Below the threshold the estimates are much smaller and statistically indistinguishable from zero. Above the threshold, acquirers patent more than 6% less than their controls on average over the five years, and targets nearly 5% less. By the fifth year the gap is wider still, about 14% for acquirers and 8% for targets. The decline therefore comes from the acquisitions that give the acquirer a stake of some size in the target's profit, which is what an internalized externality predicts.⁴⁵

⁴²Panels A and B of Table I, with log-point coefficients of -0.059 and -0.026 , and of -0.052 and -0.020 .

⁴³7.0 for acquirers and 6.9 for targets, with small pre-acquisition coefficients (Figure A.XVI in the Appendix).

⁴⁴The stake is the share held after the transaction, which for about two thirds of the acquisitions is the stake acquired in the transaction itself and otherwise includes an earlier holding. A 10% cutoff is a common threshold for disclosure and for the exemption of passive investments from merger review. Splitting the acquisitions above 10% further at 25% gives similar estimates in both groups.

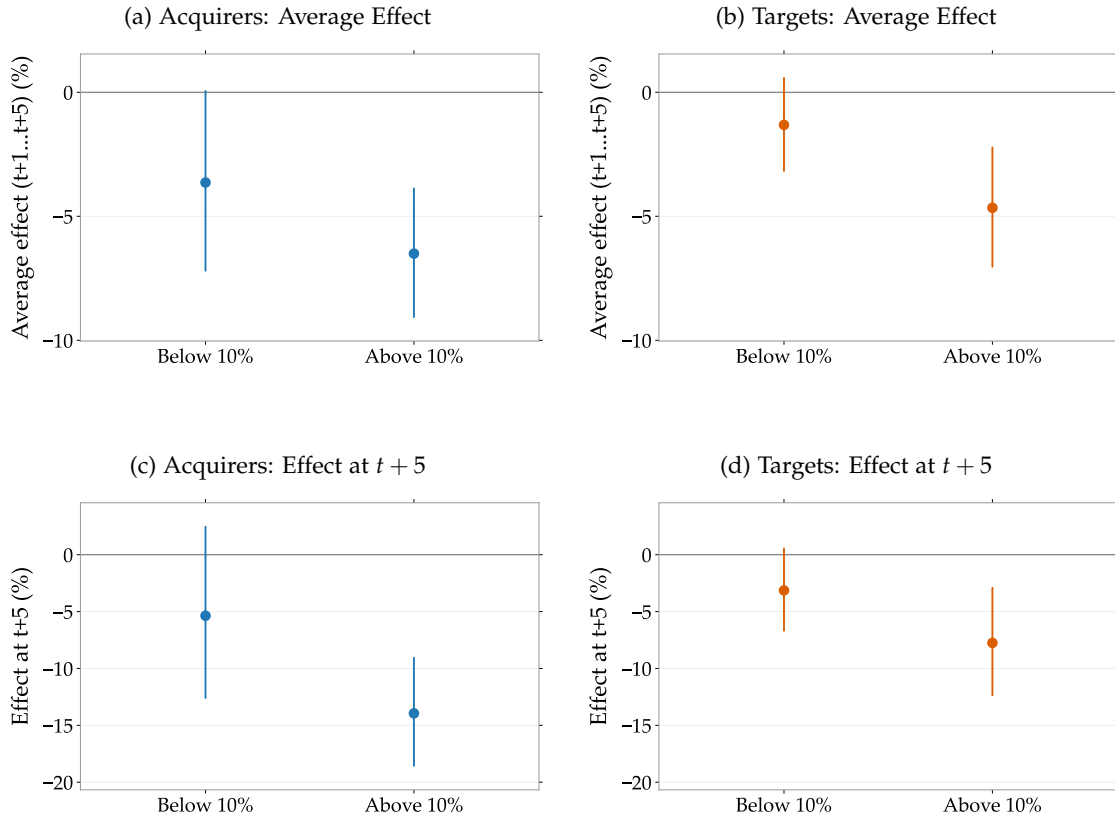
⁴⁵Predictions P1 and P2 of Section 3. Figure III reports the split.

Table I: *Estimated Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output*

	Acquirers	Targets
	(1)	(2)
Panel A: Binary treatment		
Treatment effect	−0.059*** (0.012)	−0.026*** (0.008)
Panel B: Discrete treatment		
Treatment effect	−0.052*** (0.011)	−0.020*** (0.007)
Observations	24,038	57,927

Each column reports the estimated average effect over the five-year period post-acquisition from a staggered difference-in-differences regression using the de Chaisemartin and D'Haultfoeuille (2026) estimator. The results represent weighted averages of group-time effects. Panel A treats the acquisition as binary (0 before an overlapping minority stake acquisition, 1 thereafter). Panel B treats the treatment as discrete (0 before, 1 after the first deal, 2 after the second, etc.). The event-study figures report effects in percent, $100 \times (\exp(\beta) - 1)$, whereas the table reports log-point coefficients. All specifications include firm and year fixed effects and use a 5-year post-acquisition window. Standard errors are clustered at the firm level and shown in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Figure III:** *Effects on Patents by the Size of the Stake*

The figure shows the estimated average effect on cumulative patents over the five post-acquisition years (top panels) and the effect five years after the acquisition (bottom panels) for acquisitions with a stake held below and above 10% (278 and 641 acquirers, 1,304 and 765 targets). Control firms enter each group with the weight of their matched treated firms in that group. Binary treatment as in Panel A of Table I; points are the estimates, bars the 95% confidence intervals based on standard errors clustered at the firm level.

6.2 Assessing the Credibility of the Main Estimates

The pre-acquisition coefficients in Figure II are flat, but flat estimated pre-trends do not by themselves rule out violations of parallel trends. Small pre-existing differences could bias the post-acquisition estimates in either direction (Roth et al., 2023; Rambachan and Roth, 2023).

If a pre-existing difference simply continued after the acquisition, removing that difference should remove the effect. Following Fenizia and Saggio (2024), we fit a linear trend to the pre-acquisition coefficients and extrapolate it into the post-acquisition period (left panels of Figure IV). For acquirers the estimated slope is essentially zero, and rotating the event-study coefficients around the extrapolated trend (right panels) leaves the estimates virtually unchanged. For targets the fitted slope is -0.0021 log points per year, so the rotation shifts the post-acquisition path toward zero. The rotated estimates remain negative and significant from the second year onward.

The rotation is a descriptive device that does not account for the estimation uncertainty of the fitted trend. The Rambachan–Roth analysis below provides the corresponding formal inference. At $M = 0$ its robust confidence intervals are equivalent to removing an estimated linear trend while fully accounting for that uncertainty. For $M > 0$ they additionally allow for non-linear deviations.

The “honest” difference-in-differences approach of Rambachan and Roth (2023) bounds the permissible change in the slope of the differential trend between treated and control firms:

$$\Delta\text{Slope Differential} = \{\theta : |(\theta_{k+1} - \theta_k) - (\theta_k - \theta_{k-1})| \leq M\}.$$

The parameter M controls how much the slope of the differential trend is allowed to bend in each post-acquisition period. The effect remains identified as long as the counterfactual trend deviates from a straight line by at most M percentage points per period. At $M = 0$ an arbitrary *linear* differential trend is permitted. The pre-acquisition coefficients then determine its slope, though only up to estimation error, and the procedure removes that line from the post-acquisition estimates while widening the interval by the uncertainty in the slope. The rotation above removes the same line, but treats its slope as known. We report the fifth post-acquisition coefficient, where cumulative deviations from linearity are largest and the test is most demanding.

Allowing for arbitrary linear violations of parallel trends ($M = 0$), the patent effects at $t + 5$ remain clearly bounded away from zero on both sides.⁴⁶ The same holds for the average effect over the five-year window. Both effects thus survive when the entire pre-acquisition divergence is attributed to a continuing trend rather than to the acquisition.

⁴⁶The robust confidence intervals at $t + 5$ are $[-12.1\%, -3.9\%]$ for acquirers and $[-6.8\%, -1.9\%]$ for targets. The average effect over the five years survives as well, at $[-6.5\%, -1.8\%]$ and $[-3.9\%, -1.1\%]$.

Beyond that, the counterfactual slope may steepen by a further 0.19 (acquirers) and 0.14 (targets) percentage points in every post-acquisition period before the year-five intervals include zero. For targets, each year of permitted bending amounts to about two thirds of the entire fitted pre-acquisition slope. By year five the counterfactual slope may steepen to more than three times the fitted slope. For acquirers, whose fitted pre-trend is essentially flat, the permitted deviations are large relative to anything estimated in the pre-period. The same sensitivity analysis applied to citation-weighted output yields the same conclusion.⁴⁷

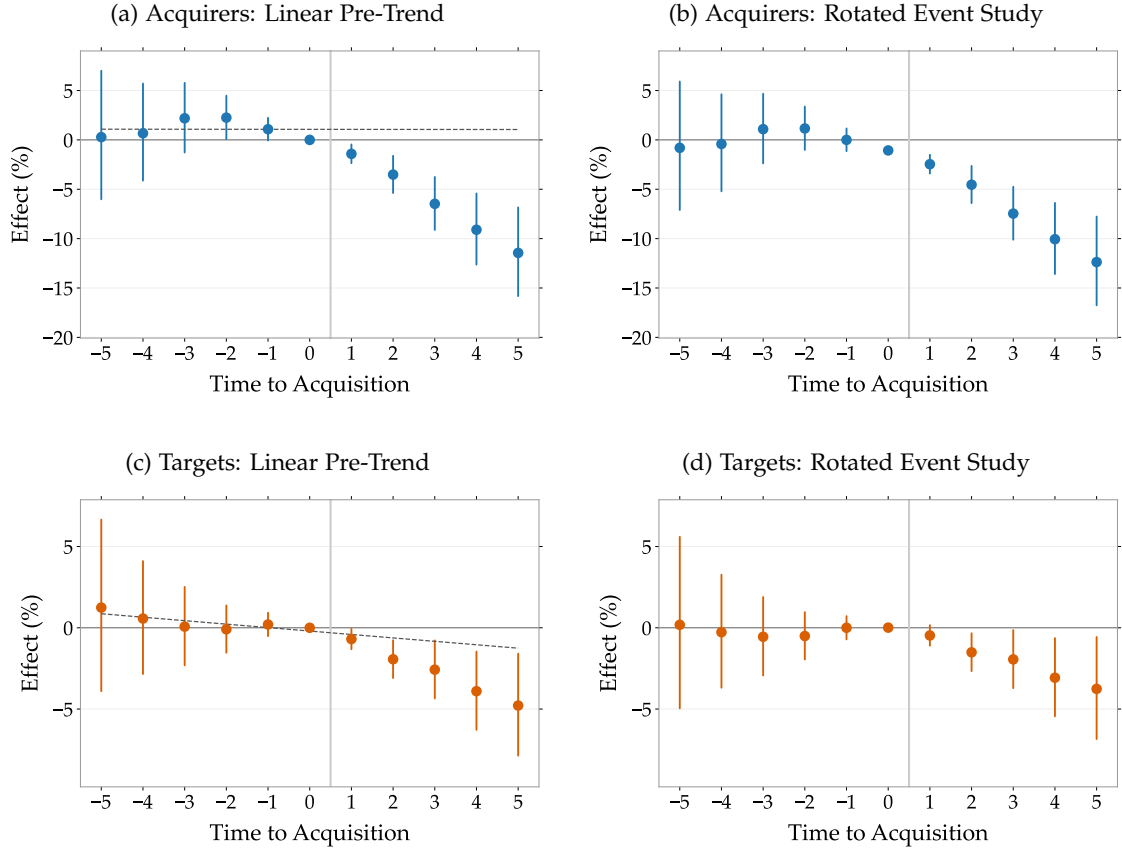


Figure IV: Pre-Trend Extrapolation and Rotated Event Studies (Patents)

The left panels show the patent event studies together with a linear trend (dashed line) fitted by OLS to the pre-acquisition coefficients (five years before the minority stake acquisition, including the reference period) and extrapolated into the post-acquisition period. The right panels rotate the event-study coefficients around this extrapolated trend, re-anchored at the reference period. Confidence intervals (95%, standard errors clustered at the firm level) in the right panels are those of the original event-study coefficients, shifted with the rotation; they do not incorporate the estimation uncertainty of the fitted trend, which is addressed formally by the Rambachan–Roth sensitivity analysis in the text.

The confidence intervals in Figure II cover each coefficient individually with 95% probability, but not the path of all ten coefficients jointly. A pre-acquisition path that looks flat coefficient by coefficient need not be flat as a whole. Following the recommendation of Baker et al. (2025), we compute sup- t simultaneous confidence bands (Olea and Plagborg-Møller, 2019), which cover

⁴⁷ At $M = 0$ the robust confidence intervals at $t + 5$ for citation-weighted output are $[-13.5\%, -4.2\%]$ for acquirers and $[-7.7\%, -1.2\%]$ for targets. Figure A.VII in the Appendix reports the intervals for a grid of M values.

the entire coefficient path with 95% probability.⁴⁸ Figure V shows the corresponding widening of the bands. No pre-acquisition coefficient reaches the simultaneous threshold, so the pre-acquisition path is jointly indistinguishable from zero. The post-acquisition decline survives, from the first year onward for acquirers and from the second for targets, and the year-five effects of both groups remain bounded away from zero.

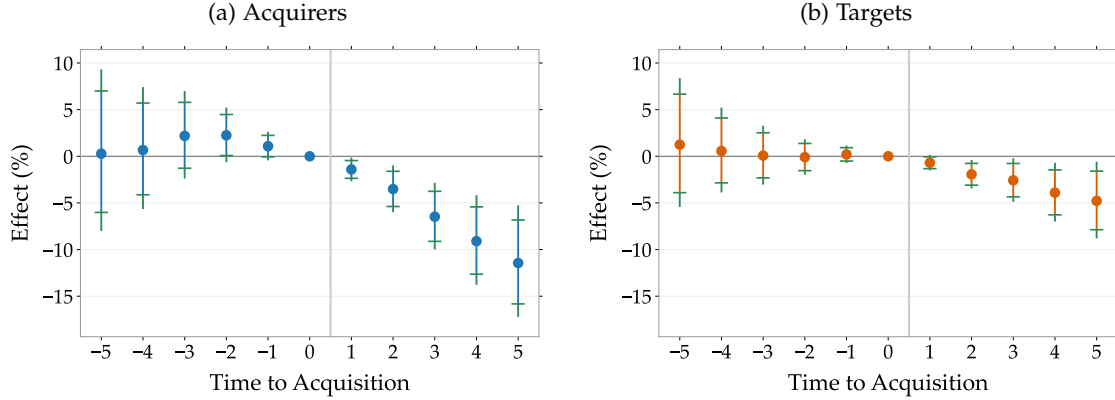


Figure V: Sup-t Simultaneous Confidence Bands (Patents)

The figure repeats the patent event studies of Figure II. The inner (light) bars are the pointwise 95% confidence intervals; the green extensions widen them to sup-t simultaneous 95% confidence bands (Olea and Pлагborg-Møller, 2019), which cover the entire path of event-study coefficients jointly with 95% probability. Critical values are computed from 200,000 draws from the estimated correlation matrix of the coefficients. Standard errors are clustered at the firm level.

6.3 Citation-Weighted Innovative Output

One question is whether the decline is a loss of research or only of filings. A count treats every filing alike, so it falls either way. Protection costs money, in every jurisdiction and over the life of the patent. A firm trimming that budget drops the applications it values least. The acquisitions would then have cut filings the firms could spare rather than research, and the decline would not be the loss of innovation our argument takes it to be. If the retreat were confined to marginal, low-value patents, citation-weighted output should fall by less than the count.

We therefore re-estimate the event studies with the log of cumulative forward citations as the dependent variable. Figure VI reports the result. Citation-weighted output falls by about as much as the patent count, 14% for acquirers and 5% for targets by the fifth year. The cutback is therefore not confined to marginal, low-value patents. The acquisitions reduced the amount of patenting rather than its composition.

⁴⁸The resulting critical values, obtained from 200,000 draws of the estimated correlation matrix of the coefficients, are 2.61 (acquirers) and 2.56 (targets) instead of the pointwise 1.96.

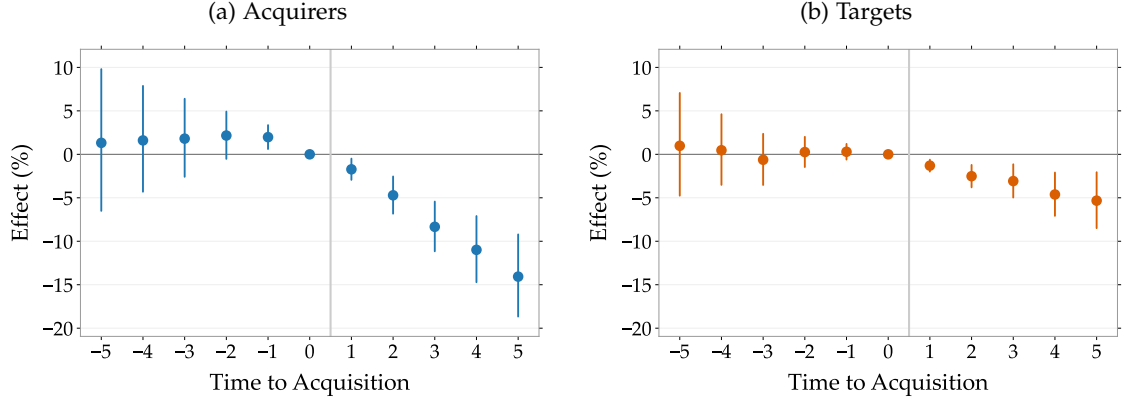


Figure VI: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Citation-Weighted Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of forward citations to the firm's granted patents, per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

6.4 Does Technological Overlap Matter?

Without technological overlap, the acquirer's innovation has little scope to take profit from the target, so there is little externality to internalize.⁴⁹ A minority stake should then leave the innovation of both parties unchanged. Minority acquisitions between firms that do not share any 4-digit IPC class therefore serve as a placebo. We repeat the full matching procedure for these firms, drawing their controls from firms that were never party to a minority stake or a majority acquisition. We assign treatment at the firm's first such acquisition, and exclude firms that also experience an overlapping acquisition in that year or the following five.

Figure VII plots the event study. The coefficients are small throughout the post-acquisition window and show none of the widening gap of Figure II, and no post-acquisition coefficient is statistically significant for either side. Table II puts the average effects at -0.2% for acquirers and -0.8% for targets, neither statistically significant. A minority stake by itself does not reduce patenting. The decline arises only between firms whose technologies overlap.

⁴⁹Prediction P3 of Section 3.

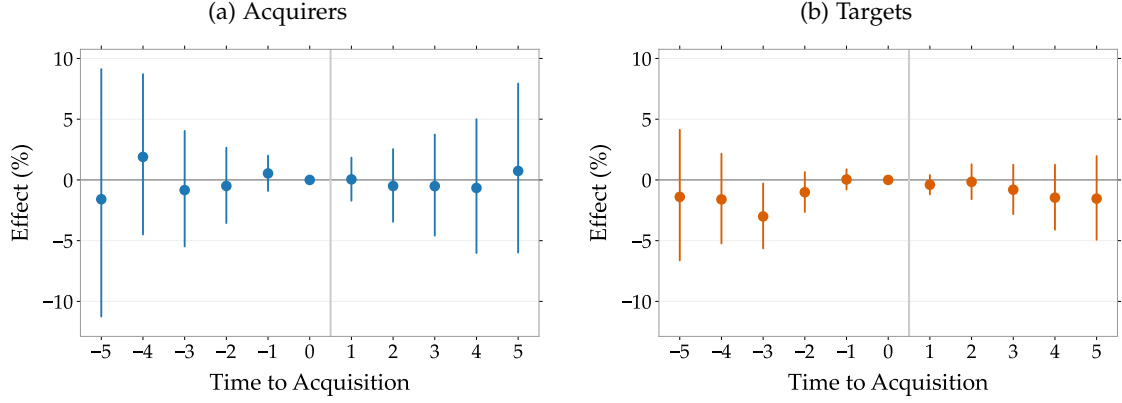


Figure VII: Event Study on the Effects of Non-Overlapping Minority Stake Acquisitions on Cumulative Patent Output

The figure presents coefficient estimates in percentage for each of the five years before and five years after the non-overlapping minority stake acquisition of treated firms. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

Table II: Estimated Effects of Non-Overlapping Minority Stake Acquisitions on Cumulative Patent Output

	Acquirers	Targets
	(1)	(2)
Treatment effect	−0.002 (0.019)	−0.008 (0.009)
Observations	10,141	34,822

Each column reports the estimated average effect over the five-year period post-acquisition from a staggered difference-in-differences regression using the de Chaisemartin and D'Haultfoeuille (2026) estimator. The results represent weighted averages of group-time effects. The acquisition is treated as binary (0 before a non-overlapping minority stake deal, 1 thereafter). All specifications include firm and year fixed effects and use a 5-year post-acquisition window. Standard errors are clustered at the firm level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 Innovation Inside and Outside the Shared Technology Space

The externality operates where the two firms' technologies meet. If the decline reflects softer R&D competition, it should be concentrated in the classes the firms share rather than spread evenly across their portfolios.⁵⁰

To see whether it is, we exploit variation *within* each firm's technology portfolio. For each acquisition, we classify a firm's technology areas into *overlapping* classes, those 4-digit IPC classes where both the acquirer and target held at least one patent prior to the acquisition, and *non-overlapping* classes, where they do not. The classification errs on the side of caution. Since 4-digit IPC classes are narrow, some technologically related classes end up in the non-overlapping group, biasing against finding a difference.

⁵⁰Non-overlapping technology classes may also be affected if they are complements or substitutes to the overlapping technologies.

We then collapse each firm's patenting into two components, the cumulative number of patents in overlapping classes and in non-overlapping classes.⁵¹ We estimate the event study of equation (8) separately for each, with the same estimator and treatment definition as in the firm-level analysis. The overlapping component is measured on the same set of classes for both firms, namely the treated firm's overlapping classes, which are fixed before the acquisition. Because the control firm contributes its patenting in these very classes, a shared field that loses momentum for reasons unrelated to the acquisition slows both series alike and differences out of the comparison. The non-overlapping component covers each firm's patenting in all other classes.⁵²

In the classes the two firms share, patenting falls on both sides (Figure VIII). By year five, acquirers have accumulated 11.4% and targets 5.5% fewer patents in these classes than the control group. The gap opens in the first year after the acquisition and widens steadily thereafter. Outside the shared classes, the two sides differ. Targets keep patenting exactly as their controls do, and the two series never separate. Acquirers also patent somewhat less outside the shared classes, but the gap is about half the size and statistically insignificant until the fifth year. For targets, the decline is confined to the technologies the two firms have in common. For acquirers, it is steepest there.

⁵¹A patent with several IPC classes is assigned to the overlapping component if at least one of its classes is an overlapping class and to the non-overlapping component otherwise, so the two components sum to the firm-level total.

⁵²For this analysis we additionally require control firms to have a similar pre-acquisition composition of their patent portfolio. The share of a control firm's pre-acquisition patenting that falls into its treated counterpart's overlapping classes must be within 10 percentage points of the treated firm's own share. Each component is estimated on the firms that were active in the respective classes two years before the acquisition.

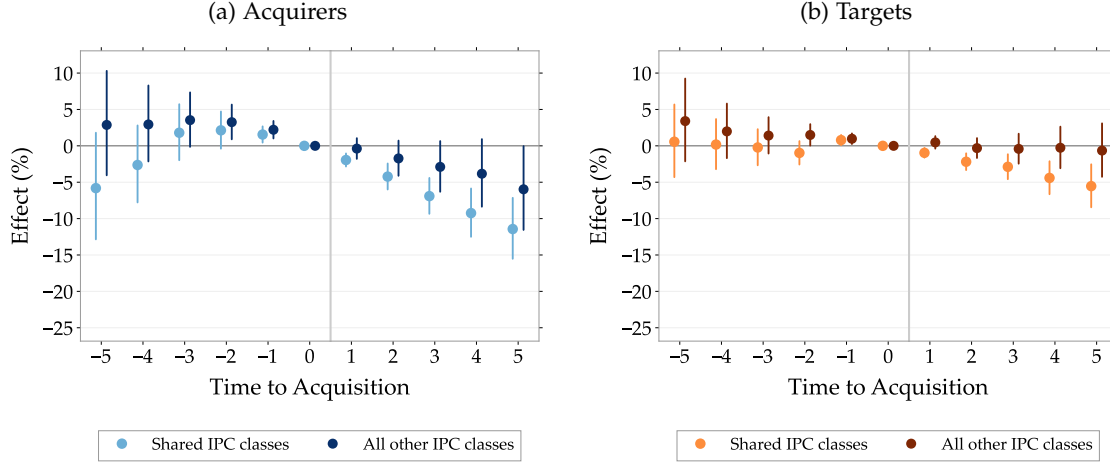


Figure VIII: Event Studies for Shared and All Other Technology Classes

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for the firm's shared IPC classes (light markers) and all of its other IPC classes (dark markers). The dependent variable is the log of the cumulative number of granted patents (dated by application year) in the respective set of classes, per firm and year. The overlapping component is measured on the same pre-acquisition set of overlapping IPC classes for the treated firm and its matched control; the non-overlapping component comprises each firm's own remaining IPC classes. Each component is estimated on the firms that held patents in the respective set of classes two years before the acquisition. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

This pattern is consistent with the internalization of the externality. Once the acquirer holds a stake in its rival, competing along the shared technological trajectory pays less, and the target, whose returns in these classes move with the acquirer's effort, follows.

The decline in the shared classes could in principle reflect a change in patenting rather than in innovation. Once the rival holds a stake, patents filed mainly to block that rival or to deter litigation lose part of their purpose, and firms may protect more of their work through secrecy. Joint research between the parties would, if anything, show up as joint filings. These are rare and do not increase. Fewer than five percent of the overlapping acquisitions in our sample ever produce a joint patent, and only 1.2 percent do so in the five years after the deal. Again, citation-weighted output falls by about as much as the patent count, so the decline is not confined to filings of little technological value.⁵³

6.6 When Does the Target Respond?

In the model, the target does not react to the deal itself but to the acquirer's cutback.⁵⁴ It does so only to the extent that its own returns depend on what the acquirer does in the shared classes. Our definition of overlap is deliberately coarse. A single shared 4-digit IPC class is enough to classify an acquisition as overlapping, and Figure I shows how much this hides. For almost a

⁵³Section 6.3.

⁵⁴Prediction P2 of Section 3.

third of the targets the shared classes hold only a few percent of their patents, for nearly a fifth they hold virtually all of them. The shared classes may thus be the core of a firm's research or a marginal sideline, and the two parties need not be equally exposed.

The externality requires exposure on both sides. The acquirer's innovation can only depress the target's profit to the extent that the target actually earns in the contested classes. The target, in turn, can only respond to the extent that its own returns move with what the acquirer does there. We therefore measure, for each party, the share of its pre-acquisition patents that falls into the shared classes, and take the smaller of the two. This "shared-space importance" is high only when the contested classes are central for *both* firms. It is low whenever they are peripheral for either one.

We then split the sample at the median of this measure across deals, 16%. One group holds the deals in which the shared classes account for an above-median share of both firms' pre-acquisition patenting, the other the rest.⁵⁵ The outcome here is again total patenting at the firm level, as in the baseline specification of Table I, and not one of the two portfolio components of the previous figures. What varies across the two groups is the kind of deal, not the set of classes we count.

Figure IX shows that the two sides respond to different conditions. For targets, the effect is concentrated in the deals where the shared space matters to both parties. The year-five effect is -8.7% in that group ($n = 639$), while the remaining deals show no measurable response at any horizon ($n = 921$).⁵⁶ For acquirers, by contrast, the split makes no difference. Both groups decline significantly, by 10–13% by year five, bracketing the full-sample estimate of Figure II. The target thus moves only where the shared space matters to both sides, which is where a reaction to the acquirer's behavior should show up. The acquirer cuts back regardless.

⁵⁵We apply the same cutoff to acquirers and targets. Side-specific medians differ considerably (10% for targets, 25% for acquirers) and would classify the same deal as material for one party but immaterial for the other, mechanically generating asymmetries between the two sides.

⁵⁶The two groups comprise fewer deals than the matched sample because the measure is undefined whenever the counterpart cannot be identified in the deal data or holds no pre-acquisition patents; such deals cannot be classified and are excluded from the split.

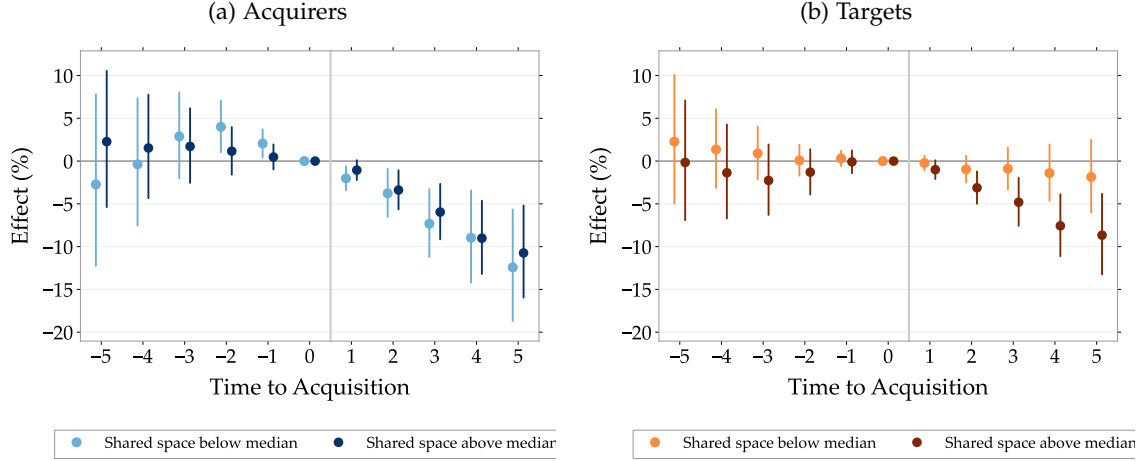


Figure IX: Event Studies by the Importance of the Shared Technology Space for Both Parties

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for two groups of deals. For each deal we compute the share of each party's pre-acquisition patents that falls into the classes the two firms share, and take the smaller of the two shares. Dark markers show deals for which this share is above its sample median of 16%, i.e. the shared classes account for an above-median share of both firms' pre-acquisition patenting. Light markers show all remaining deals. Control firms enter each group with the weight of their matched treated firms in that group. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

The median cutoff is not a knife edge. In Figure A.VIII in the Appendix we vary the cutoff from 10% to 50% of both portfolios. The pattern is the same throughout. The year-five effect for the above-cutoff target group lies between -7% and -9% at every cutoff. The below-cutoff group stays between -2% and -4% and is statistically indistinguishable from zero, except at the 50% cutoff, where it comprises most of the sample. Acquirers decline significantly in both groups at every cutoff. What separates the two groups of targets is thus not how much technology the two firms share, but whether the technology they share is material to both of them.

This also puts our headline estimates in perspective. Because the binary overlap definition counts any shared class, the target sample mixes deals with no measurable response together with deals with a year-five cumulative gap of about nine percent. We retain the binary definition as our main specification, because it is transparent and can be applied to every deal. The average target effect we report therefore understates the response of the firms whose research programs genuinely collide.

Product-market overlap adds a second dimension. The externality is strongest when the two firms not only research in the same fields but also sell in the same markets. The target's response should then be larger when the deal parties are product-market rivals as well. Figure X splits the technologically overlapping acquisitions by whether the direct deal parties also share a four-digit NACE industry code. For targets, the year-five estimate is -8.5% with product-market overlap and -3.3% without. The difference between the two is imprecisely measured, but it points the

same way as the split by shared-space importance. The target's response is largest where the two firms meet as competitors. For acquirers, the estimates are negative and of similar size in both groups.⁵⁷

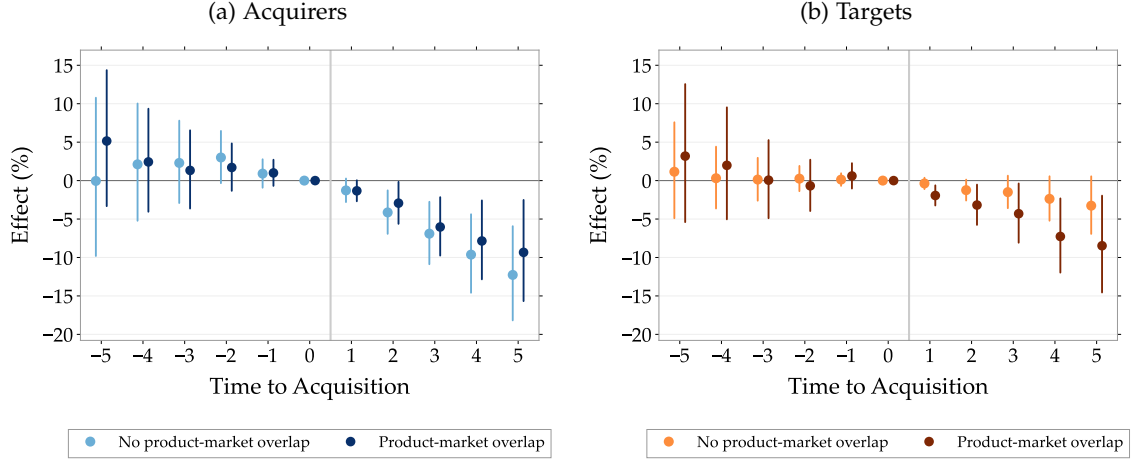


Figure X: Event Studies by Product-Market Overlap of the Deal Parties

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms, estimated separately for two groups of deals. Product-market overlap means that the direct deal parties share at least one four-digit NACE code. Deals for which the industry code of one party is unavailable (12% of targets and 28% of acquirers) cannot be classified and are excluded from the split. Control firms enter each group with the weight of their matched treated firms in that group. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. The estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each series shows 95% confidence intervals based on standard errors clustered at the firm level.

Taken together, the four patterns point in one direction. The decline appears only when the two firms' technologies overlap. Both parties cut back, the cutback is concentrated in the classes they share, and the target's response is strongest where the shared space matters to both firms and where they also compete in product markets.

Firm-wide explanations such as financial constraints or managerial distraction are difficult to reconcile with the first three of these patterns. The elimination of duplicate research efforts would more naturally produce substitution or specialization across the two parties than parallel cutbacks concentrated in the fields they share. Within the individual classes a pair shares, the two parties cut back together rather than dividing the work between them. The classes in which the acquirer cuts back most are the classes in which the target cuts back most.⁵⁸ A governance channel does not fit either. The acquirer could use board seats or information rights to influence

⁵⁷The pattern is not specific to NACE. Defining industry overlap through four-digit SIC or NAICS codes, or combinations of SIC, NACE, and NAICS, yields five-year target estimates between -8.5% and -10.1% with product-market overlap and between -2.7% and -3.3% without, and acquirer estimates between -9.0% and -9.4% and between -11.9% and -12.8% , respectively.

⁵⁸For each acquisition we take the 4-digit IPC classes in which both parties filed at least one patent in the five years before the deal. For each side we measure the change from those five years to the five years after as the difference in filings divided by their sum. Regressing the target's change on the acquirer's across the 5,224 shared classes of 495 acquisitions, with acquisition fixed effects, gives a slope of 0.12 with a 95% confidence interval of 0.07 to 0.17. Removing each class's own average movement leaves 0.07. Table A.V in the Appendix reports the full set of specifications. A division of the shared classes between the two parties would require a negative slope.

the target's research, but that would not produce a decline concentrated in the shared classes and absent without technological overlap.

A selective reorganization of research along the technological relationship cannot be fully separated from weakened competitive incentives with our data. Only the internalization of the externality predicts that the target responds where the shared space matters to both firms.

6.7 Alternative Samples and Definitions of Overlap

The evidence so far rests on one sample and on one definition of technological overlap.

Excluding Japan and Korea. Minority shareholdings play a special institutional role in Japan and Korea. Cross-held minority stakes are a defining feature of the *keiretsu* business groups and “stable shareholder” relationships in Japan and of the *chaebol* conglomerates in Korea. In these environments a minority stake may reflect group affiliation and relational governance rather than a strategic acquisition of a stake in a rival. The two countries together account for roughly half of the overlapping acquisitions in our sample. If the results came from these environments, excluding them should remove the effect.

We therefore re-estimate the model excluding all treated firms, and their matched controls, from Japan, from Korea, and from both countries jointly. Figure XI reports the average effects over the five-year window and the year-five effects on patents across the four samples. Every exclusion leaves both the average and the year-five effects statistically significant for both groups. If anything, the effects strengthen when Japan is excluded. To the extent that cross-held stakes within business groups are less likely to reflect a strategic acquisition of a stake in a rival, including them dilutes the estimated effect.

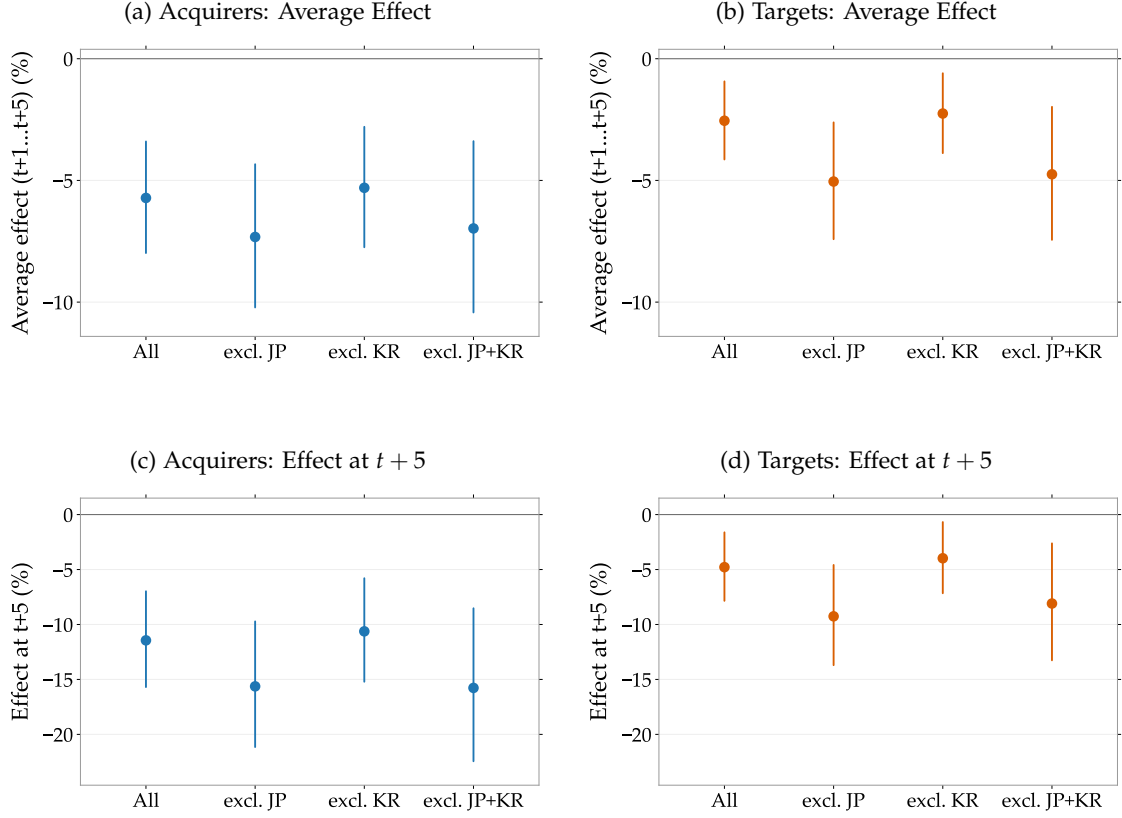


Figure XI: Effects on Patents When Excluding Japan and Korea

The figure shows the estimated average effect on cumulative patents over the five post-acquisition years (top panels) and the effect five years after the acquisition (bottom panels) in four samples: the full sample, excluding Japan, excluding Korea, and excluding both. Each exclusion drops all treated firms headquartered in the respective country together with their matched control firms. Points are the estimates, bars the 95% confidence intervals based on standard errors clustered at the firm level.

Spillovers to Control Firms. One concern is that the comparison itself may be contaminated by the deal. The matched control firms patent in the same 4-digit IPC classes as the treated firms. If the acquisition changes competitive conditions in these fields, such technologically close controls may themselves respond, and the comparison would then partly reflect that response rather than the behavior of unaffected firms. We address this by re-matching treated firms to controls that are active in the same 3-digit IPC classes but share no 4-digit class with the treated firm. These controls remain technologically comparable while standing outside the directly affected fields. The acquirer effect is essentially unchanged under this matching, with a year-five effect on patents of -11.6% (Figure A.XII in the Appendix). For targets, the smaller pool of admissible controls leaves the estimate imprecise.

Escalation to Majority Control. The baseline sample excludes acquisitions in which the acquirer obtains majority control of the target within five years, so that the estimates are not confounded by outright takeovers. A second concern is that this exclusion conditions the sample on a post-acquisition event. If the deals we drop are the ones whose patenting behaves differently,

putting them back should move the estimates. We therefore re-run the analysis retaining these deals, censoring the affected firms from the year the stake turns into a majority position, so that their minority-stake years remain in the panel. The matched sample grows to 2,128 targets and 1,004 acquirers. Both estimates move slightly toward zero, to -10.3% for acquirers and -3.5% for targets, and both remain significant (Figure A.XIII in the Appendix).

Overlap Defined by Industry Codes. Our main specification defines overlap by proximity in the technology space. As an alternative, we classify an acquisition as overlapping if the acquirer and target share at least one 4-digit NACE industry code, and match on industry codes rather than IPC classes. The estimates remain negative and significant, reaching -4.8% for acquirers and -5.9% for targets by year five (Figure A.XV in the Appendix). For targets the magnitude is comparable to the technology-based estimate, for acquirers it is roughly 40 percent of it. This is consistent with the mechanism operating through technological overlap. The externality runs through innovation in overlapping technologies, so technological overlap should be the sharper screen for innovation responses, with product-market overlap a coarser proxy.

The results are robust to the exclusion of Japan and Korea, to controls drawn from outside the affected technology classes, to the reinclusion of acquisitions that escalate to majority control, and to the industry-code definition of overlap. An artifact of the sample or of the overlap definition is unlikely to survive all four. In each case, a firm that acquires a stake in a technologically overlapping target cuts back its research in the technologies the two of them share, and the target follows. By contrast, neither happens when the two firms work on different technologies.

7 Conclusion

Firms may compete less aggressively through innovation once one firm owns part of another. When the acquirer and target operate in overlapping technology fields, the acquirer bears part of the loss that its own innovation imposes on the target. That weakens its incentive to innovate against it. The target may then respond to the acquirer's pullback by reducing its own innovation.

Consistent with this mechanism, both acquirers and targets patent less after minority stake acquisitions between such firms. By the fifth post-acquisition year, cumulative patent output is roughly 11% lower for acquirers and 5% lower for targets than for their matched controls. Citation-weighted output follows a similar path. We find no comparable decline after minority stake acquisitions between firms whose technologies do not overlap.

The decline is closely related to where the two firms innovate. For acquirers and targets alike, the decline is concentrated in the technology classes the two firms share. Minority ownership

therefore affects not only how much firms patent, but also where they patent.

Acquirers and targets respond under different conditions. Targets cut back mainly when the shared technology space matters to both firms and when the two firms also compete in product markets. That is what one would expect if the target reacts to a change in the acquirer's behavior rather than to the deal itself. Acquirers cut back regardless. Taken together, these patterns point to weakened competitive incentives rather than to financial constraints or a division of research between the two parties.

Partial acquisitions often receive less scrutiny than full mergers, and every transaction in our sample stays below majority ownership, with most stakes below 25%. They are common, accounting for roughly a quarter of global M&A activity by deal count, and the firms that enter them are large and research-intensive. The findings are therefore relevant for competition policy. They do not imply that every minority acquisition warrants closer review. We find no comparable effect when the firms' technologies do not overlap, and reviewing every minority investment would impose substantial costs on agencies and firms.

Technological overlap can therefore help identify the transactions in which innovation incentives are most likely to change. The importance of the shared fields and the product-market overlap between the parties show where target responses are strongest. More broadly, our results show that minority ownership can reduce innovation in the technologies two firms share even when the stake remains far below a majority position.

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A Appendix

A.1 Sample Construction and Representativeness

Table A.I details how we constructed the datasets for acquirers and targets that we use in the paper and the number of firms that remained after each step. To ensure that our selected group of firms constitutes a representative sample, we compare our group of treated firms with the entire set of companies involved in minority stake acquisitions as reported in the Zephyr dataset. The distribution of our final sample across years is similar to other firms involved in minority stake acquisitions (see Figure A.I). The later years (2018–2019) are under-represented in our sample because we require that treated firms have at least three years of consecutive patent data. Figure A.II shows the distribution of our final sample across countries. The distribution of our final sample across industries is also similar to other firms involved in minority stake acquisitions (see Figure A.III). Similar to other firms, our sample of acquirers and targets predominantly operate in the manufacturing sector. As expected, our sample is under-represented in terms of Finance, Insurance, and Management companies, since we excluded non-corporate firms and investment offices from the group of treated firms.

A.2 Figures

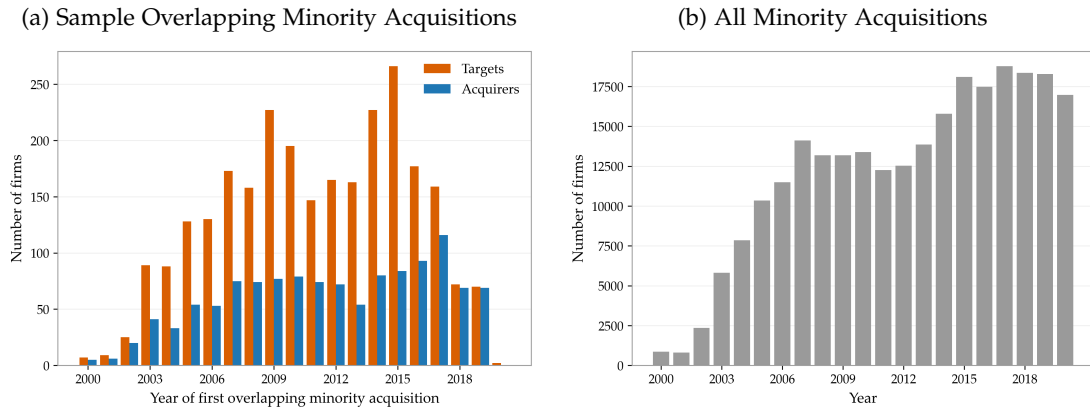


Figure A.I: *Distribution of Minority Stake Acquisitions Over Time*

Panel (a) shows the distribution of the number of targets and acquirers in our sample involved in overlapping minority stake acquisitions by year, over the sample period. Panel (b) shows the distribution of all firms involved in any minority stake acquisition by year for the period from 2000 to 2020, as per the Zephyr database.

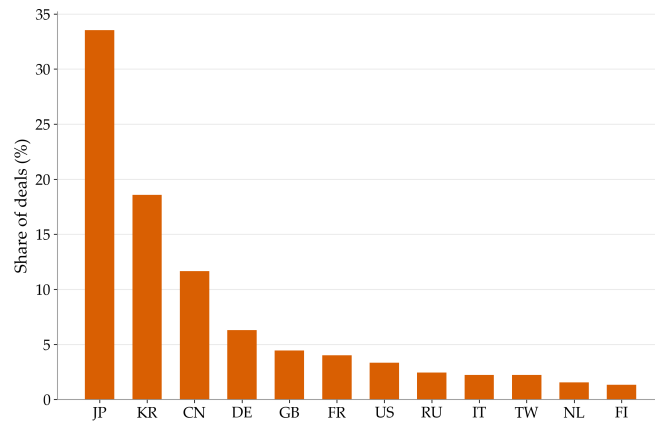


Figure A.II: *Distribution of the Sample Across Countries*

The figure shows the distribution of the overlapping minority stake deals in our sample by the country of the treated firm, pooling targets and acquirers.

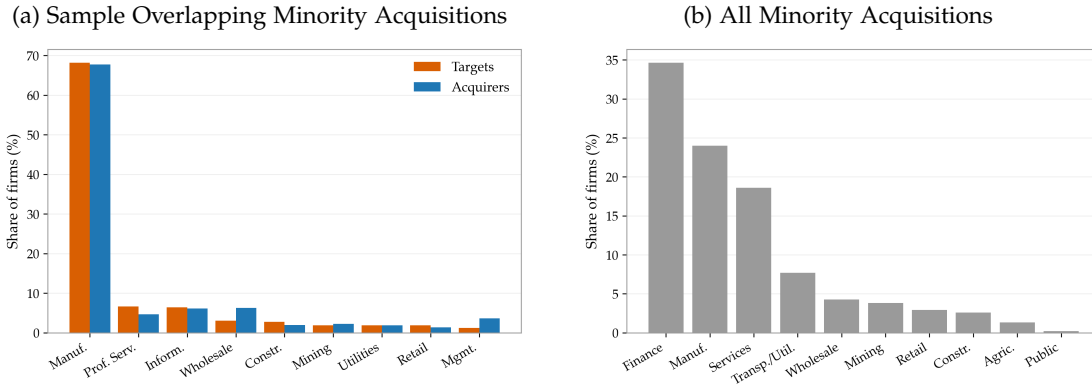


Figure A.III: Distribution of Minority Stake Acquisitions Across Industries

Panel (a) shows the distribution of the number of targets and acquirers in our sample involved in overlapping minority stake acquisitions by sector (based on 2-digit NAICS codes). Panel (b) shows the distribution of all firms involved in any minority stake acquisition by sector (based on the primary US SIC code reported in the Zephyr database).

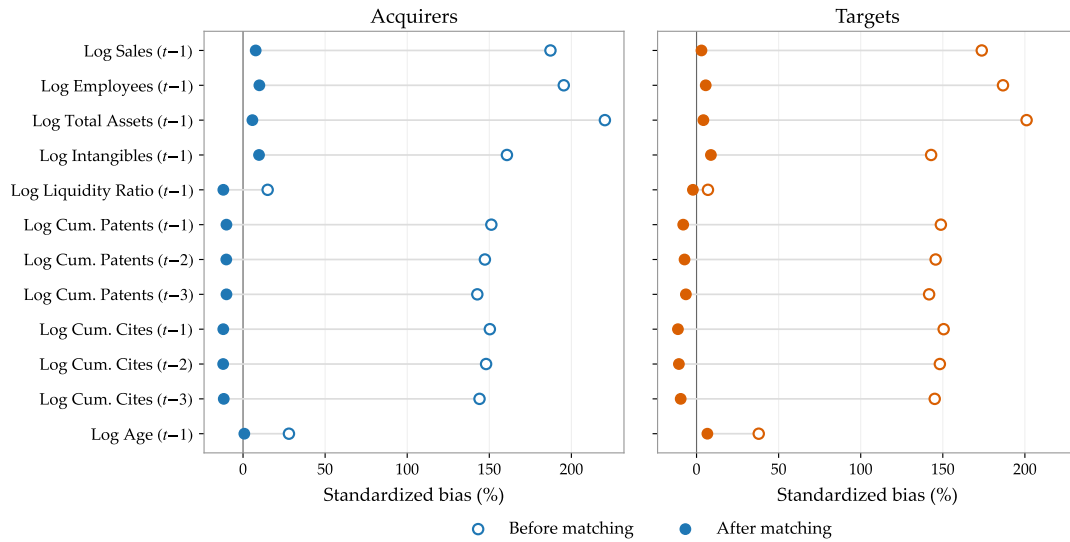


Figure A.IV: Standardized Biases, Before and After Matching

This figure plots the standardized biases for the covariates used in the propensity score estimation, before and after matching, separately for targets and acquirers. For each regressor, the standardized bias is calculated as the difference of sample means in the treated and control samples as a percentage of the square root of the average of sample variances in both groups (Caliendo and Kopeinig, 2008). Before matching, biases are computed using all firm-year observations from the pool of potential controls alongside the $t-1$ observation for treated firms. After matching, biases are computed using the $t-1$ observation for matched treated and control firms only, with control firms weighted by the number of treated firms they serve. The matched firms are then followed as a panel for the event-study estimation.

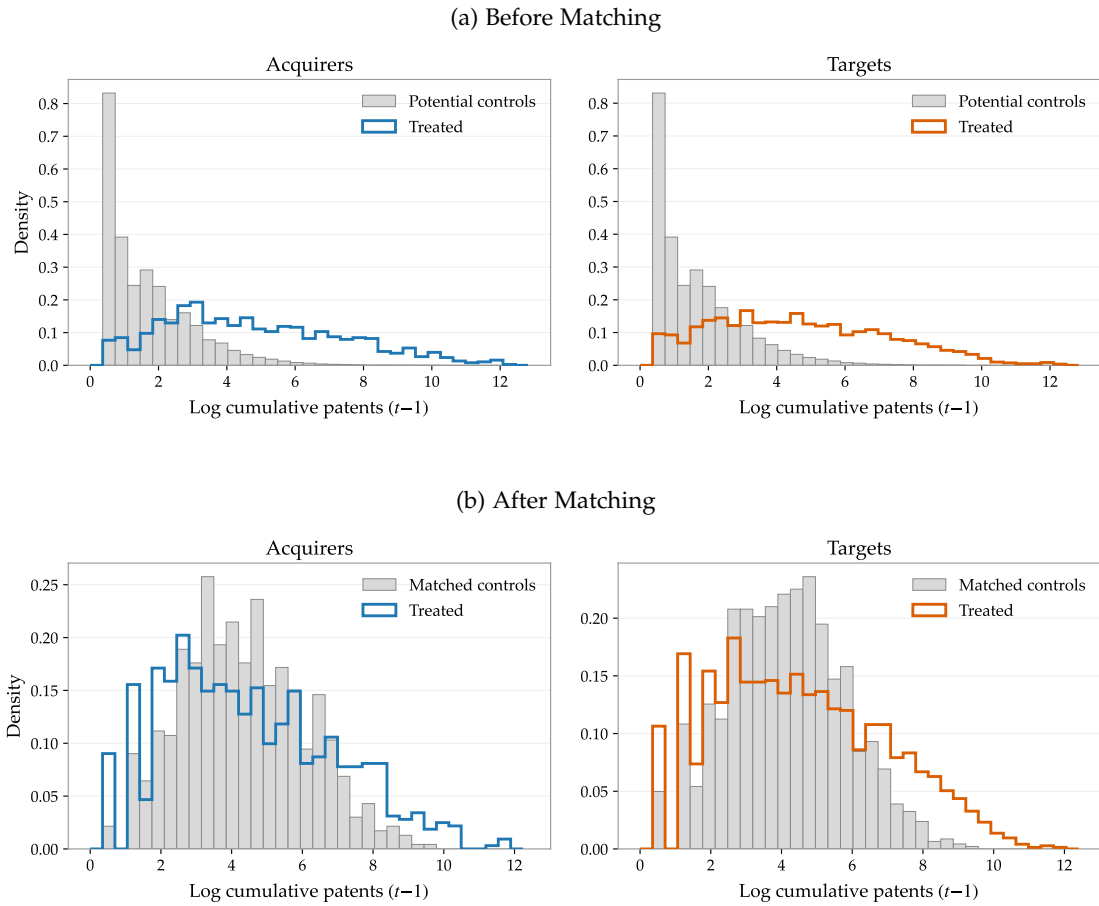


Figure A.V: Distribution of Patents for Treated and Control Firms, Before and After Matching

The top panels plot the distribution of the log of the cumulative number of patents for treated firms (the year before the acquisition) and all potential control firms. The bottom panels plot the distribution of the log of the cumulative number of patents for the matched treated and control firms (the year before the acquisition).

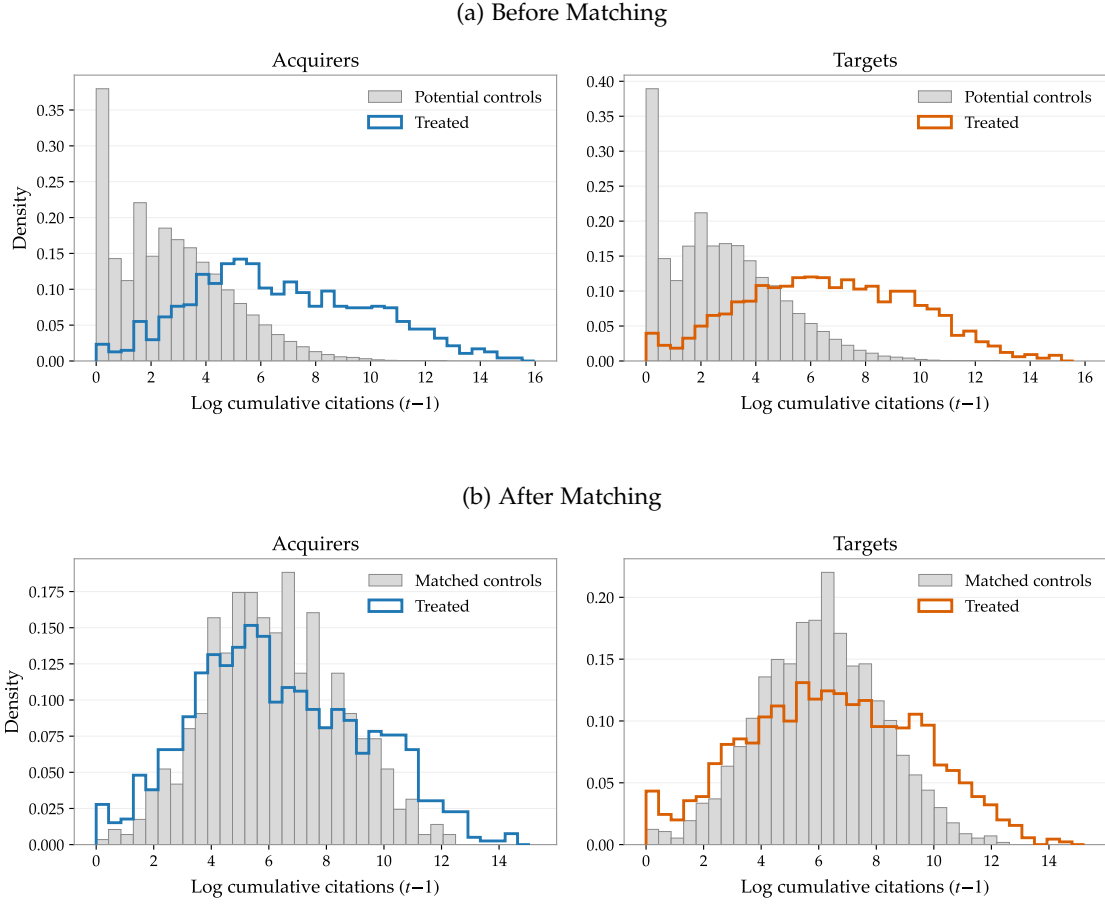


Figure A.VI: *Distribution of Patent Citations for Treated and Control Firms, Before and After Matching*

This figure compares the distribution of patent citations for treated and control firms, before and after the matching. The top panels plot the distribution of the log of the cumulative number of patent citations for treated firms (the year before the acquisition) and all potential control firms. The bottom panels plot the distribution of the log of the cumulative number of patent citations for the matched treated and control firms (the year before the acquisition).

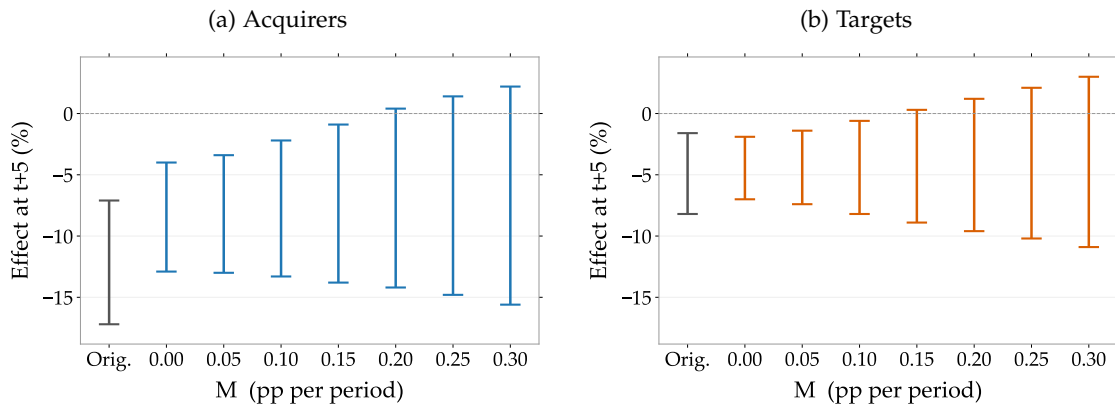


Figure A.VII: *Robust Confidence Intervals for the Year-Five Patent Effect Under Bounded Violations of Parallel Trends*

The figure reports Rambachan and Roth (2023) robust 95% confidence intervals for the effect on cumulative patents five years after the overlapping minority stake acquisition, as a function of the sensitivity parameter M (horizontal axis, in percentage points per period). M bounds the period-to-period change in the slope of the differential trend between treated and control firms: $M = 0$ allows an arbitrary linear differential trend, and larger values additionally allow non-linear deviations from linearity. “Orig.” denotes the original event-study confidence interval. The robust intervals are fixed-length confidence intervals computed from the event-study coefficients with five pre-acquisition (placebo) coefficients. Standard errors are clustered at the firm level.

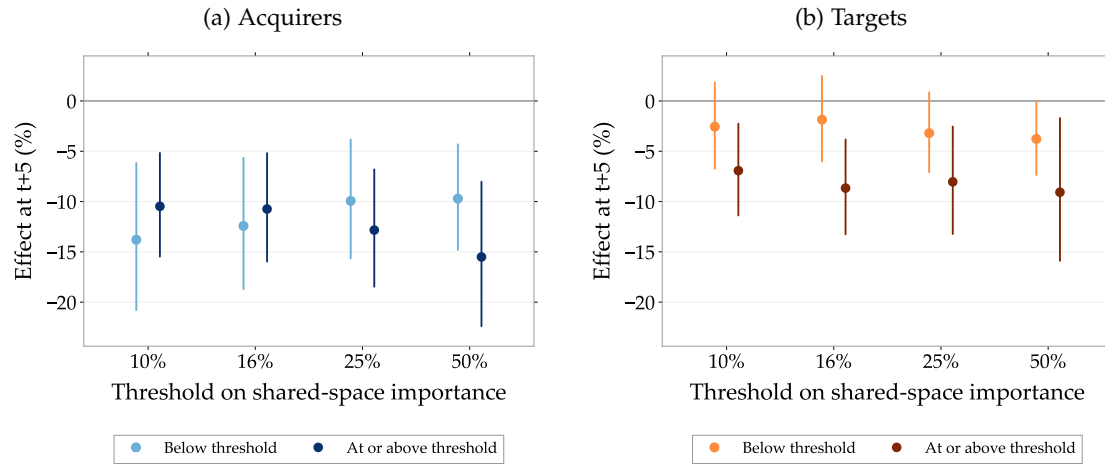


Figure A.VIII: Year-Five Effects by the Threshold on Shared-Space Importance

The figure reports the estimated effect on cumulative patents five years after the acquisition, separately for deals above and below a threshold on the importance of the shared technology space, for four different thresholds. For each deal we compute the share of each party's pre-acquisition patents that falls into the classes the two firms share and take the smaller of the two shares; the threshold is applied to that measure and is identical for acquirers and targets. Each treated firm's matched control is assigned to the same group. Group sizes for acquirers (below/above) are 253/666, 346/573, 460/459 and 655/264; for targets 813/747, 921/639, 1057/503 and 1269/291. The mean of the measure in the group above the threshold rises from 0.46 to 0.75 across the four cutoffs, and in the group below it from 0.03 to 0.14. Estimates use the estimator presented in de Chaisemartin and D'Haultfoeuille (2026) with firm and year fixed effects; bars are 95% confidence intervals based on standard errors clustered at the firm level.

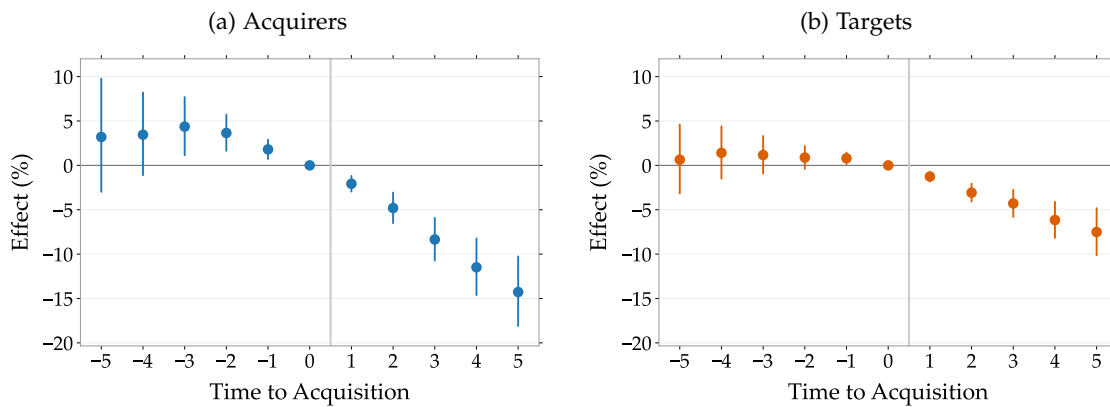


Figure A.IX: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Unweighted Estimation

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, the estimation does not weight control firms by the number of treated firms they serve as a match for; the matched sample is identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

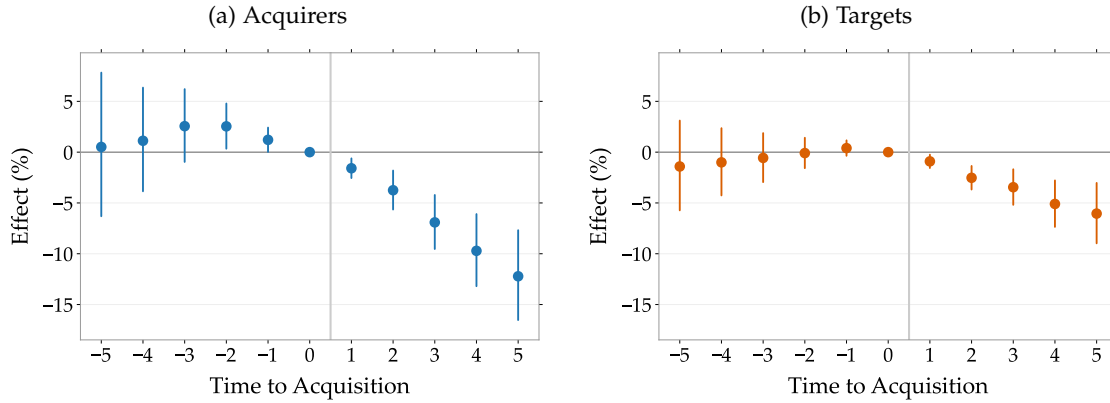


Figure A.X: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching Without Replacement

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, the matching is performed without replacement, so that each control firm serves as the match for at most one treated firm; all other elements of the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

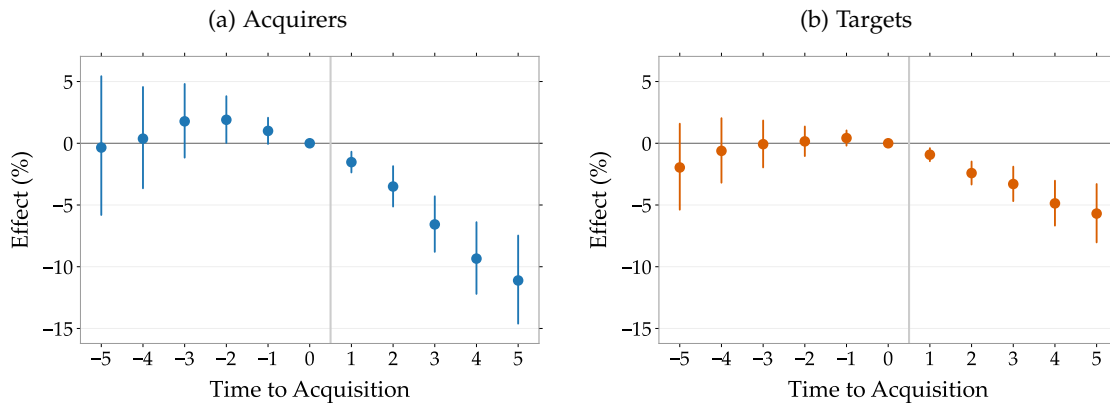


Figure A.XI: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching on Three Nearest Neighbors

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, each treated firm is matched to its three nearest neighbors in terms of the propensity score instead of one; all other elements of the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

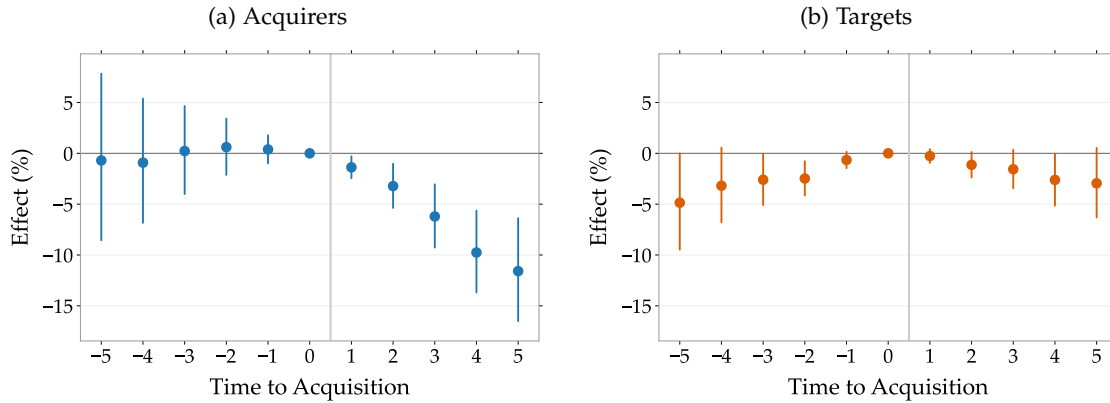


Figure A.XII: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Matching on 3-digit IPC Overlap Excluding 4-digit Overlap

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, the control group is matched on 3-digit IPC overlap while excluding firms that share any 4-digit IPC class with the treated firm. This ensures that control firms operate in the same broad technology area but their technologies do not overlap those of the treated pair. All other elements of the matching procedure (propensity score covariates including the lagged innovation levels, calipers, and common support) are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

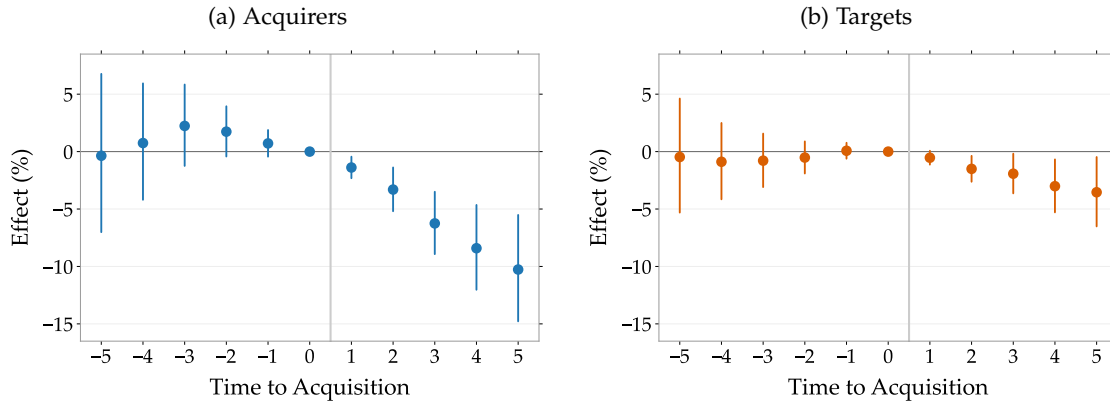


Figure A.XIII: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Retaining Deals that Escalate to Majority Control

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, deals in which the acquirer obtains majority control of the target within five years of the minority acquisition are retained; the affected firms are instead censored from the year the stake turns into a majority position. All other elements of the sample construction and the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

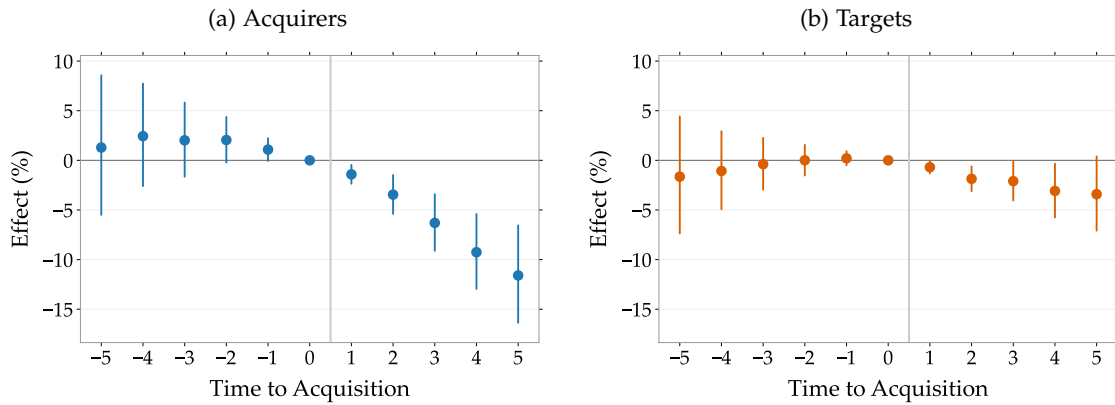


Figure A.XIV: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Censoring at the Second Overlapping Acquisition

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. In contrast to the baseline event study in Figure II, treated firms are censored from the year of their second overlapping minority stake acquisition onward, so that the estimates reflect exposure to the first acquisition only; all other elements of the sample construction and the matching procedure are identical to the baseline. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

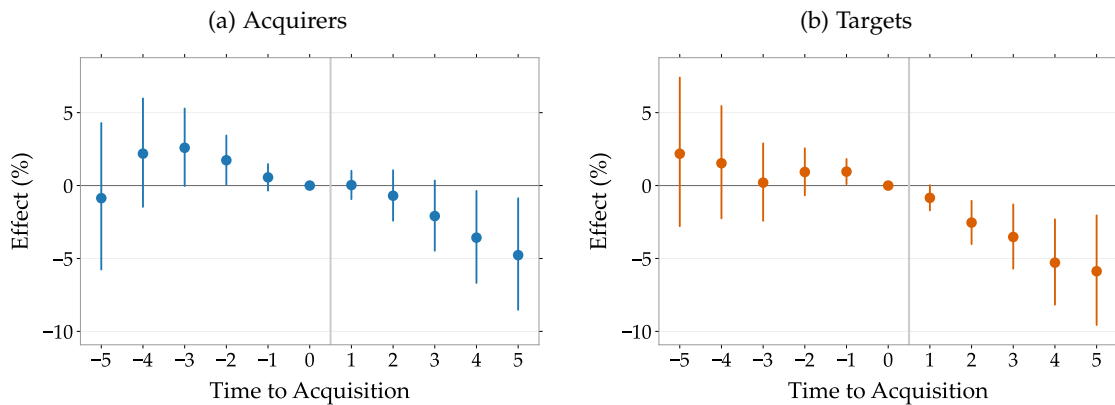


Figure A.XV: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Cumulative Patent Output – Overlap Defined by Industry Codes

The figure presents coefficient estimates in percentage for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. Overlapping deals are defined by an overlap in the 4-digit NACE industry codes of the target and its acquirer/parent; the matching procedure likewise requires industry-code overlap instead of IPC overlap. The dependent variable is the log of the cumulative number of granted patents (dated by application year), per firm and year. This estimation uses the estimator presented in de Chaisemartin and D'Haultfoeuille (2026). Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

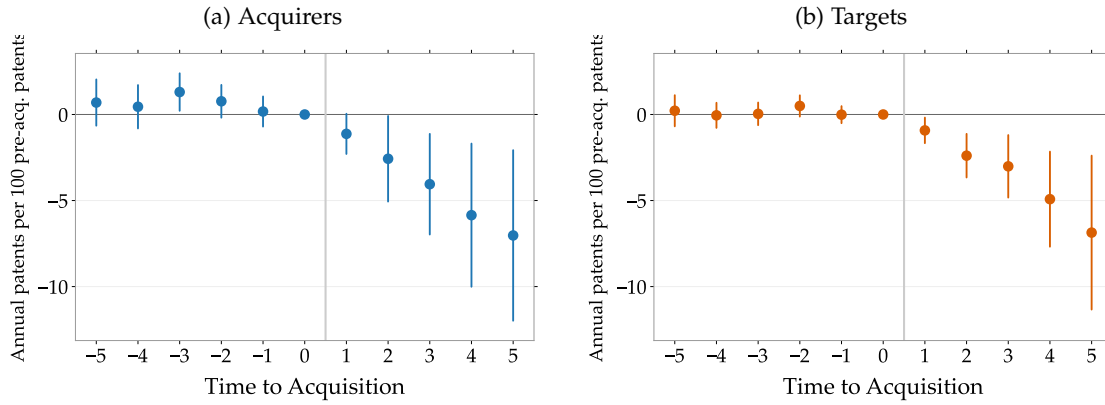


Figure A.XVI: Event Study on the Effects of Overlapping Minority Stake Acquisitions on Annual Patent Production

The figure presents coefficient estimates for each of the five years before and five years after the overlapping minority stake acquisition of treated firms. The dependent variable is the number of granted patents (dated by application year) filed by the firm in year t , divided by one plus the firm's cumulative number of granted patents in the last pre-acquisition year, expressed per 100 pre-acquisition patents. Estimates use the same matched sample and the estimator presented in de Chaisemartin and D'Haultfoeuille (2026), with each control firm aligned to the acquisition cohort of its matched treated firm. Event time $\tau = 1$ is the acquisition year and $\tau = 0$, the last pre-acquisition year, serves as the reference period. Time fixed effects and firm fixed effects are included. Each panel shows the 95% confidence intervals based on standard errors clustered at the firm level.

A.3 Tables

Table A.I: Dataset Construction

	Action	Remaining Companies	
		Targets	Acquirers
Step 1	Firms involved in overlapping minority stake acquisitions	4,561	2,898
Step 2	Eliminate deals in which the acquirer obtains majority control of the target within 5 years of the minority acquisition	4,345	2,734
Step 3	Eliminate firms without Orbis financial data	2,783	1,311
Step 4	Eliminating non-corporate firms (e.g., financial company, mutual and pension fund)	2,730	1,261
Step 5	Eliminate firms without 3 years of consecutive patent data	2,682	1,239

This table details how we constructed the datasets for targets and acquirers that we use in the paper and the number of firms that remained after each step.

Table A.II: Propensity Score Estimation (Probit Model)

	Targets	Acquirers
	(1)	(2)
Log Sales ($t-1$)	-0.186 (0.014)***	-0.160 (0.019)***
Log # Employees ($t-1$)	0.226 (0.013)***	0.130 (0.018)***
Log Total Assets ($t-1$)	0.266 (0.014)***	0.341 (0.019)***
Log Intangibles ($t-1$)	0.044 (0.003)***	0.053 (0.005)***
Log Liquidity Ratio ($t-1$)	0.141 (0.020)***	0.249 (0.026)***
Log Cum. Patents ($t-1$)	0.139 (0.102)	0.227 (0.144)
Log Cum. Patents ($t-2$)	-0.044 (0.150)	-0.435 (0.207)**
Log Cum. Patents ($t-3$)	0.009 (0.089)	0.329 (0.121)***
Log Cum. Patent Citations ($t-1$)	0.111 (0.065)*	-0.051 (0.110)
Log Cum. Patent Citations ($t-2$)	0.011 (0.085)	0.303 (0.129)**
Log Cum. Patent Citations ($t-3$)	-0.053 (0.054)	-0.205 (0.068)***
Log Age ($t-1$)	-0.097 (0.015)***	-0.022 (0.022)
Observations N	641,737	655,122
Pseudo R^2	0.41	0.45

This table estimates the probability of being involved in an overlapping minority stake acquisition. The dependent variable equals one if the firm was involved in such a deal in the following period. The probit model uses all firm-year observations from the pool of potential controls, but only pre-deal observation for treated firms. For treated firms with more than one overlapping minority acquisition, matching is performed on the year before the first acquisition. Variables with zero values are transformed as $\log(\text{variable} + 1)$. The lagged levels of cumulative patents and citations at $t-1$, $t-2$, and $t-3$ are included to match treated and control firms on the shape of their pre-acquisition innovation trajectory. The three levels of the same cumulative stock are highly collinear, so their individual coefficients are not informative in isolation. Country, year, and industry (2-digit NAICS) fixed effects are included. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.III: Pre-Acquisition Summary Statistics

(a) Acquirers

	Mean	10-Perc.	Median	90-Perc.	Obs.
<i>Sales(t-1)</i>	4.6	0.0	0.5	13	1,171
<i>Number of Employees(t-1)</i>	16,184	137	2,111	40,009	1,171
<i>Age(t-1)</i>	37	8.0	23	82	1,171
<i>Fixed Assets(t-1)</i>	3.7	0.0	0.3	10	1,167
<i>Total Assets(t-1)</i>	6.2	0.1	0.8	18	1,171
<i>Intangible Fixed Assets(t-1)</i>	0.8	0.0	0.0	0.9	1,151
<i>Liquidity Ratio(t-1)</i>	2.3	0.6	1.2	4.2	1,156
<i># Granted Patent Applications(t-1)</i>	155	0.0	5.0	230	1,167
<i>Cumulative # Granted Patent Applications(t-1)</i>	3,174	4.0	69	3,871	1,167
<i># Patent Cites(t-1)</i>	2,441	0.0	18	2,519	1,167
<i>Cumulative # Patent Cites(t-1)</i>	51,645	13	458	57,516	1,167

(b) Targets

	Mean	10-Perc.	Median	90-Perc.	Obs.
<i>Sales(t-1)</i>	2.6	0.0	0.4	6.7	2,543
<i>Number of Employees(t-1)</i>	10,193	105	1,568	23,613	2,543
<i>Age(t-1)</i>	40	7.0	30	85	2,543
<i>Fixed Assets(t-1)</i>	2.0	0.0	0.2	4.7	2,539
<i>Total Assets(t-1)</i>	3.4	0.0	0.4	8.1	2,543
<i>Intangible Fixed Assets(t-1)</i>	0.6	0.0	0.0	0.6	2,528
<i>Liquidity Ratio(t-1)</i>	2.0	0.6	1.2	3.6	2,533
<i># Granted Patent Applications(t-1)</i>	76	0.0	3.0	129	2,543
<i>Cumulative # Granted Patent Applications(t-1)</i>	1,786	3.0	62	2,913	2,543
<i># Patent Cites(t-1)</i>	1,008	0.0	10	1,272	2,543
<i>Cumulative # Patent Cites(t-1)</i>	27,049	10	530	33,752	2,543

This table provides summary statistics on the acquiring and target firms included in our sample. The table summarizes the values of firm-level variables for acquirers and targets in the pre-acquisition period. The Sales and the Assets are expressed in billion euro. The Liquidity Ratio is the ratio of current assets to current liabilities. Values are measured in the year before the acquisition. Outliers are excluded from this table by dropping the top percentile of observations in terms of Sales(t-1).

Table A.IV: Relative Size of Parties Pre-Acquisition**(a) Acquirers**

	10-Perc.	25-Perc.	50-Perc.	75-Perc.	90-Perc.	Obs.
<i>Ratio of Sales</i>	0.32	1.05	4.79	18.05	69.57	742
<i>Ratio of # of Employees</i>	0.28	0.95	3.42	13.80	43.34	745
<i>Ratio of Age</i>	0.36	0.74	1.15	2.08	5.33	743
<i>Ratio of Fixed Assets</i>	0.36	1.14	5.12	26.05	101	741
<i>Ratio of Total Assets</i>	0.40	1.20	4.75	18.59	64.32	746
<i>Ratio of Intangible Fixed Assets</i>	0.03	0.49	3.90	33.35	209	671
<i>Ratio of Liquidity</i>	0.30	0.56	0.93	1.62	3.18	732
<i>Ratio of # Granted Patents</i>	0.00	0.01	1.04	8.00	35.10	457
<i>Ratio of Cumulated # Granted Patents</i>	0.08	0.50	3.38	18.50	144	743
<i>Ratio of # Patent Cites</i>	0.00	0.00	1.00	14.00	79.39	433
<i>Ratio of Cumulated # Patent Cites</i>	0.05	0.37	3.37	29.27	220	723

(b) Targets

	10-Perc.	25-Perc.	50-Perc.	75-Perc.	90-Perc.	Obs.
<i>Ratio of Sales</i>	0.01	0.03	0.12	0.56	1.98	915
<i>Ratio of # of Employees</i>	0.01	0.05	0.18	0.80	2.69	915
<i>Ratio of Age</i>	0.19	0.47	0.86	1.33	2.64	914
<i>Ratio of Fixed Assets</i>	0.00	0.02	0.12	0.63	2.15	911
<i>Ratio of Total Assets</i>	0.01	0.03	0.14	0.64	1.98	915
<i>Ratio of Intangible Fixed Assets</i>	0.00	0.01	0.09	1.06	7.70	848
<i>Ratio of Liquidity</i>	0.34	0.61	1.03	1.81	3.40	902
<i>Ratio of # Granted Patents</i>	0.00	0.00	0.04	0.50	3.24	643
<i>Ratio of Cumulated # Granted Patents</i>	0.00	0.03	0.28	2.01	17.22	912
<i>Ratio of # Patent Cites</i>	0.00	0.00	0.01	0.44	3.99	630
<i>Ratio of Cumulated # Patent Cites</i>	0.00	0.02	0.20	1.92	17.55	891

Panel (a) compares the acquirers and their respective targets across several variables the year before the minority stake acquisition. Similarly, Panel (b) compares the targets and their respective acquirers across several variables the year before the minority stake acquisition. This table presents the distribution (percentiles) of the ratios of values between targets and their acquirers (and between acquirers and their targets) for different variables in the pre-acquisition period (t-1). Observations for which (i) there are more than two parties in the acquisition, (ii) there are missing data for the other party in the acquisition, are excluded from this table. Outliers are excluded from this table by dropping the top percentile of observations in terms of Ratio of Sales.

Table A.V: Co-movement of the Two Parties Across Their Shared Technology Classes

	Slope	95% CI	Shared classes	Acquisitions	Groups absorbed
<i>Panel A. At least one pre-acquisition filing on each side</i>					
No fixed effects	0.149	[0.092, 0.206]	5,595	866	–
Acquisition fixed effects	0.122	[0.070, 0.174]	5,224	495	–
+ technology class	0.072	[0.027, 0.118]	5,224	495	0
+ technology class $\times$ five-year period	0.061	[0.015, 0.106]	4,878	490	346
+ technology class $\times$ acquisition year	0.057	[0.008, 0.107]	3,881	474	1,343
Mean within-acquisition correlation	0.122	[0.090, 0.153]		237	–
<i>Panel B. At least three pre-acquisition filings on each side</i>					
No fixed effects	0.151	[0.081, 0.221]	2,073	363	–
Acquisition fixed effects	0.152	[0.072, 0.231]	1,929	219	–
+ technology class	0.073	[0.004, 0.142]	1,929	219	0
+ technology class $\times$ five-year period	0.069	[–0.010, 0.148]	1,698	210	231
+ technology class $\times$ acquisition year	0.179	[0.089, 0.269]	1,136	196	793
Mean within-acquisition correlation	0.164	[0.111, 0.216]		116	–

For each overlapping minority stake acquisition we take the 4-digit IPC classes in which both parties filed at least one patent in the five years before the acquisition. For each side we measure the change from those five years to the five years after as the difference in filings divided by their sum, which lies between -1 and $+1$ and makes large and small classes comparable without further adjustment. The dependent variable is the target's change in a class and the regressor is the acquirer's change in the same class, where the acquirer side comprises the acquirer and its immediate corporate parents, as in our definition of an overlapping acquisition. Standard errors are clustered by acquisition. The final column reports the number of observations absorbed because their fixed-effect group contains a single shared class and therefore carries no within-group variation. It is large for the class-by-acquisition-year specification, which is why that estimate is the least informative of the five. The final row of each panel is not a regression. It reports the correlation between the two sides' changes computed separately within each acquisition and then averaged, over the acquisitions that share at least four classes. A division of the shared classes between the two parties, with one withdrawing where the other continues, would require a negative slope.